

**Council of Scientific & Industrial Research Fourth Paradigm Institute
Bengaluru, Karnataka**

**Analysis of Convective and Stratiform Rainfall Structures in Monsoon Depressions
over the Indian Subcontinent Using GPM IMERG and Ancillary Datasets (2014–2024)**

Summer Internship And Research

Submitted By:

Aditya Roy

224ER1005

National Institute of Technology

Rourkela - 769008

Submitted To:

Dr. Bhimala Kanta Rao

Principal Scientist

CSIR Fourth Paradigm Institute

Bengaluru, Karnataka - 560037

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DECLARATION

I, Aditya Roy, declare that this report titled Analysis of Convective and Stratiform Rainfall Structures in Monsoon Depressions over the Indian Subcontinent Using GPM IMERG and Ancillary Datasets (2014–2024) is my original work and has not been submitted, in whole or in parts, for any other academic or professional award. All sources of information used in this report have been duly acknowledged and referenced.

This report has been prepared as part of my academic learning under the supervision of Dr.Bhimala Kanta Rao, during the period from June to August 2025. The objective of this report is to enhance my understanding of the research process and to learn how to write a scientific paper by analyzing and interpreting existing publications.

I have made every effort to ensure that the content of this report reflects my own understanding of the topic. Any assistance received from others has been acknowledged and appropriately referenced.

(224ER1005) Aditya Roy _____

DATE: July, 2025

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This report is prepared in fulfilment of the requirement for the Atmosphere & Ocean Science course as part of the M.Tech program. I express my deepest gratitude to the Department of Earth and Atmospheric Sciences, National Institute of Technology, Rourkela, for providing me with the opportunity and resources to work on this report.

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I would like to extend my heartfelt thanks to my project supervisor and mentor, Dr. Bhimala Kanta Rao, whose guidance, insights, and support have been instrumental in shaping this work. His expertise and encouragement helped me throughout the process of analyzing and understanding the research topic.

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The experience of working on this report has enriched my technical understanding and provided a deeper insight into scientific research writing, which will undoubtedly benefit my academic and professional journey ahead.

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1. ABSTRACT

This project investigates the spatiotemporal structure and variability of rainfall associated with monsoon depressions (MDs) over the Indian subcontinent during the monsoon season (JJAS) from 2014 to 2024. The primary dataset is GPM IMERG daily rainfall, with ERA5, MEERA-2, GPM IMERG half-hourly, GPM DPR and IMD daily gauge data used for cross-validation and consistency checks.

A central component of the research is composite analysis, aligning rainfall and meteorological fields with respect to the depression center. This approach reveals the three-dimensional structure and typical distribution of rainfall, wind, temperature, and humidity during MDs, highlighting their vigorous circulation and cold-core characteristics. Composite rainfall patterns consistently show that peak rainfall is concentrated to the southwest of the depression center, with significant interannual variability in both intensity and centroid position.

Detailed radial and azimuthal analyses quantify how rainfall intensity varies with distance and direction from the MD core, capturing pronounced spatial asymmetry and dynamic evolution of precipitation fields. The study further separates convective and stratiform rainfall, demonstrating that convective rainfall is tightly confined near the center and southwest quadrant, while stratiform rainfall is more broadly distributed. Trend analysis indicates a decline in both rainfall types over central and southern India, but with localized increases in peak rain rates in the rainiest quadrants in recent years.

Overall, this work advances understanding of the rainfall structure and variability of monsoon depressions, provides a framework for evaluating precipitation datasets, and offers methodological guidance for extreme rainfall analysis and prediction across the Indian subcontinent. The findings have important implications for hydrological risk assessment, disaster preparedness, and improving monsoon forecasting.

2. INTRODUCTION

Monsoon depressions (MDs) are synoptic-scale, cyclonic, low-pressure systems that form within the monsoon trough during the summer monsoon season, most frequently over the

Bay of Bengal and occasionally over land or the Arabian Sea (Krishnamurti et al., 1975; Mishra, 2021). These systems are characterized by a broad closed cyclonic circulation, typically spanning 1,000–2,000 km horizontally and extending vertically up to 8–10 km (Krishnamurti et al., 1975; Godbole, 1977; Mishra, 2021). MDs display a cold-core structure at lower levels and a warm anomaly aloft, with the axis of the depression often tilting slightly westward or southward with height (Godbole, 1977; Mishra, 2021).

The genesis of monsoon depressions is often associated with cyclonic circulation in the mid-troposphere, and their intensification is linked to strong thermal forcing and persistent upward motion (Mishra, 2021; Diaz and Boos, 2021; Boos et al., 2014). These systems typically move northwestward or west-northwestward after formation, guided by the large-scale monsoon circulation and upper-level jet streams (Boos et al., 2014; Fletcher et al., 2018). The preferred region for their formation is the head of the Bay of Bengal, from where they travel inland, bringing widespread and intense rainfall to central, eastern, and northern India (Krishnamurti et al., 1975; Mishra, 2021).

Monsoon depressions are the principal rain-producing systems during the Indian summer monsoon, accounting for a substantial fraction of seasonal rainfall and playing a crucial role in the advancement and revival of the monsoon current (Krishnamurti et al., 1975; Mishra, 2021; Godbole, 1977). Rainfall is typically most intense in the southwest quadrant of the depression, and these systems can lead to extreme rainfall events, floods, and other hydrometeorological hazards.

The frequency and intensity of monsoon depressions are modulated by large-scale climate drivers such as the El Niño–Southern Oscillation (ENSO). El Niño years are generally associated with a reduced number of depressions and weaker monsoon rainfall, while La Niña years tend to enhance their frequency and rainfall impact (Boos et al., 2014; Adames and Ming, 2017). Other factors, including the Indian Ocean Dipole and intraseasonal oscillations, also influence the genesis and evolution of MDs.

Recent research has further clarified the structural dynamics, genesis mechanisms, and environmental interactions of monsoon depressions, using both observational and modeling approaches (Mishra, 2021; Diaz and Boos, 2021; Fletcher et al., 2018; Adames and Ming, 2017). These studies collectively underscore the central role of MDs in shaping the spatial and temporal distribution of monsoon rainfall over the Indian subcontinent.

Monsoon depressions (MDs) are responsible for a substantial share of the Indian summer monsoon rainfall. Studies estimate that MDs contribute approximately **35–55%** of the total seasonal rainfall across India, with the highest contributions (up to 55%) observed in the monsoon core zone of central India (Hunt et al., 2019; Godbole, 1977; Sikka, 2005; Francis

et al., 2016). For all-India averages, the contribution typically ranges from **27%** to nearly **50%** (Hunt et al., 2019; Sikka, 2005).

The spatial distribution of MD rainfall is highly asymmetric. The **southwest (SW) quadrant** of the depression consistently receives the most intense rainfall, with rain rates in this sector often exceeding 120 mm in 24 hours (Godbole, 1977; Hunt et al., 2016a; Vishnu et al., 2023). Central and northwest India are the wettest regions due to frequent MD tracks, while eastern and northeastern India, as well as the eastern quadrants of MDs, receive comparatively less rainfall (Vishnu et al., 2023; Romatschke and Houze, 2010).

Recent analyses reveal notable trends in the structure and intensity of MD rainfall. There is a **declining trend in area-averaged convective rainfall** over central India, but a **significant increase in peak rain rates** in the SW quadrant over the past four decades (Vishnu et al., 2023; Hunt et al., 2016a; Kumar and Pattanaik, 2023). Area-averaged stratiform rainfall from MDs also shows a **declining trend** over central India, with some increases along the west coast (Kumar and Pattanaik, 2023). The overall trend is toward more spatially concentrated and extreme rainfall in the SW sector of MDs, with less or no change elsewhere, resulting in little or no trend in storm-averaged rainfall but an increase in rainfall extremes and spatial asymmetry (Vishnu et al., 2023; Hunt et al., 2016a).

These findings highlight the critical role of monsoon depressions in the Indian monsoon system, the spatial variability of their rainfall impact, and the evolving nature of convective and stratiform rainfall associated with these systems.

A comprehensive analysis of MD-related rainfall requires the integration of multiple high-resolution precipitation datasets. The research utilizes ERA-5, MEERA-2 reanalysis products, GPM IMERG(both daily and half hour data versions) and IMD daily gauge data to assess the accuracy, consistency, and representativeness of rainfall during MD events. Special attention is given to comparing daily and half-hourly IMERG data, highlighting the advantages of sub-daily resolution for capturing rainfall variability and importance of gauge calibration for reliable daily totals.

The Global Precipitation Measurement(GPM) mission marks a major advancement in satellite-based precipitation observation, building on the foundation laid by Tropical Rainfall Measuring Mission(TRMM).The **Global Precipitation Measurement (GPM)** mission, a collaboration between NASA and JAXA, represents a major advancement in satellite-based precipitation monitoring, building upon the legacy of the Tropical Rainfall Measuring Mission (TRMM). Launched in 2014, GPM provides high-quality, near-global precipitation estimates with enhanced spatial (0.1°) and temporal (half-hourly) resolution. The core satellite is equipped with the Dual-frequency Precipitation Radar (DPR) and the GPM Microwave

Imager (GMI), enabling improved detection and classification of both convective and stratiform rainfall (Yang and Smith, 2008).

A cornerstone of GPM's data suite is the **Integrated Multi-satellitE Retrievals for GPM (IMERG)** algorithm. IMERG merges data from a constellation of international satellites, ground-based gauge networks, and reanalysis products to produce precipitation estimates at high resolution. The algorithm incorporates advanced calibration and error correction techniques, including gauge adjustment using high-density ground observations such as IMD gauge data, which improves the reliability of daily and event-based rainfall totals.

Recent studies have demonstrated that IMERG's gauge-calibrated daily and sub-daily products provide robust estimates of monsoon rainfall over the Indian subcontinent. The papers showed that IMERG daily rainfall closely matches IMD gauge observations for both mean and extreme rainfall events during monsoon depressions, outperforming earlier satellite products and most reanalyses. The sub-daily (half-hourly) IMERG data are particularly valuable for capturing the rapid evolution and diurnal cycle of rainfall associated with monsoon depressions, as highlighted by Yang and Smith (2008). These high-frequency datasets allow for detailed analysis of rainfall structure, including the separation of convective and stratiform components, and enable composite and trend studies at event, seasonal, and decadal scales.

Overall, IMERG data have become a cornerstone in modern research on Indian monsoon rainfall, supporting applications in event composites, trend analysis, hydrological modeling, and extreme rainfall attribution (Author et al., 2023).

A robust analysis of monsoon depression rainfall requires the integration of multiple high-quality datasets. In addition to the GPM IMERG satellite product, this study utilizes ERA5 reanalysis, IMD rain gauge data, and MERRA-2 reanalysis to ensure comprehensive coverage and validation.

ERA5 reanalysis, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF), provides hourly global estimates of atmospheric variables at a horizontal resolution of 0.25° . ERA5 offers physically consistent fields of precipitation, wind, temperature, humidity, and pressure, making it suitable for both event-based and climatological studies. In this work, ERA5 is used to provide synoptic context for monsoon depression events and to validate satellite rainfall estimates.

Table 1.1: Key features of GPM IMERG for monsoon depression rainfall analysis.

Feature	Description
Spatial/Temporal Resolution	0.1° grid, half-hourly to daily
Core Sensors	GMI (microwave), DPR (radar)
Convective/Stratiform Split	Enabled by multi-sensor retrievals and TRMM legacy algorithms
Gauge Calibration	Incorporates IMD and other gauge networks for improved accuracy
Validation	High agreement with IMD gauge data for MD events and monsoon extremes
Research Applications	Event composites, trend analysis, rainfall structure, hydrological modeling, extremes

The India Meteorological Department (IMD) maintains a dense network of rain gauges across the country, offering daily rainfall measurements at both station and gridded (0.25°) resolutions. IMD rain gauge data serve as the ground-truth reference for validating satellite and reanalysis rainfall products and are essential for bias correction and accuracy assessment, particularly during extreme rainfall events.

MERRA-2, produced by NASA's Global Modeling and Assimilation Office, is a global re-analysis dataset with a spatial resolution of 0.5° and temporal resolution of 1 to 3 hours. MERRA-2 provides long-term, physically consistent records of precipitation and atmospheric variables, supporting cross-validation and environmental analysis for monsoon depressions.

For the monsoon depression event from **10 October to 15 October 2020**, a comprehensive comparison was conducted using multiple precipitation datasets: GPM IMERG (daily and half-hourly), ERA5, MERRA-2, GPM DPR, and IMD rain gauge data. The objective was to determine which dataset most accurately represents the rainfall structure and evolution of this depression, with the IMD's official tracking and event chronology serving as the reference.

The datasets assessed include GPM IMERG (daily and half-hourly), ERA5 reanalysis, MERRA-2 reanalysis, GPM DPR satellite radar data, and IMD rain gauge observations. Each dataset

was evaluated for its ability to capture rainfall totals, spatial patterns, and temporal evolution over the depression's track and area of influence.

Table 1.2: Summary of supplementary datasets used in this study.

Dataset	Resolution	Main Use in Study	Strengths
ERA5	0.25°, hourly	Synoptic context, validation, composites	High-resolution, physically coherent fields
IMD Rain Gauge	Station, 0.25°, daily	Ground-truth validation, calibration	Direct observations, high spatial density
MERRA-2	0.5°, 1–3 hourly	Cross-validation, environmental analysis	Long-term, consistent fields

The IMD's official depression center positions and rainfall reports were used as the benchmark for timing, location, and intensity of rainfall maxima. Metrics such as event representation, partition accuracy, extreme rainfall accuracy, spatial resolution, and overall agreement with IMD observations were considered.

Key findings indicate that GPM IMERG daily data provided the closest match to IMD's reported rainfall patterns and depression track. It accurately captured the timing and location of peak rainfall events along the depression's path and demonstrated high spatial detail, especially for event representation and extreme rainfall detection.

ERA5 and MERRA-2 offered reasonable large-scale patterns but tended to underestimate peak intensities and showed coarser spatial gradients. These datasets were useful for synoptic context but less precise for event-scale rainfall maxima.

GPM DPR was valuable for confirming convective/stratiform structure but was limited by swath coverage and temporal gaps, making it less optimal for continuous event monitoring.

IMD rain gauge data served as the ground-truth reference for validation and was essential for calibrating and confirming the accuracy of satellite and reanalysis products.

After cross-checking all datasets against the **IMD tracking and rainfall data for the 10–15 October 2020 monsoon depression**, it was found that GPM IMERG daily data consistently provided the best outcome for representing the rainfall structure, intensity, and evolution of the event. The IMERG product’s alignment with IMD observations—both in spatial and temporal dimensions—was superior to other datasets. Other datasets (ERA5, MERRA-2, GPM DPR) were valuable for supplementary analysis and context but did not match the accuracy and detail of IMERG for this specific event.

In conclusion, for the monsoon depression of 10–15 October 2020, **GPM IMERG daily data is the most reliable and accurate dataset** for analyzing rainfall associated with the system, as confirmed by direct comparison with IMD’s official tracking and rain gauge observations. This supports the use of IMERG as the primary dataset for detailed monsoon depression rainfall studies over India.

In this study, I have utilized eleven years (2014–2024) of GPM IMERG daily precipitation data for the monsoon season (June to September, JJAS) to comprehensively analyze the rainfall characteristics of monsoon depressions over the Indian subcontinent. This long-term, high-resolution dataset serves as the primary basis for all composite analyses, trend evaluations, and structural assessments of convective and stratiform rainfall associated with monsoon depression events. The continuous coverage and fine spatial resolution of GPM IMERG daily data enable robust identification of spatial patterns, annual variability, and long-term trends in monsoon depression rainfall, as well as the construction of detailed composites and radial-azimuthal rainfall profiles.

To complement this core dataset and ensure the accuracy of event-specific analyses, I have also utilized supplementary datasets including ERA5 and MERRA-2 reanalysis products, GPM DPR radar data, and IMD rain gauge observations. These additional datasets were selectively employed for particular monsoon depression cases to provide atmospheric context, validate precipitation fields, and cross-check the accuracy of rainfall representation, especially during significant or representative events. The IMD rain gauge data, in particular, served as a ground-truth reference for validating satellite and reanalysis products. By integrating these diverse data sources, the study achieves both a comprehensive multi-year perspective and rigorous event-specific validation, leading to a more robust and nuanced understanding of the spatiotemporal structure and variability of rainfall in monsoon depressions across the Indian subcontinent.

A substantial body of research has investigated the spatiotemporal structure and variability of rainfall associated with monsoon depressions over the Indian subcontinent, with a focus on the distinction between convective and stratiform precipitation, and the application of

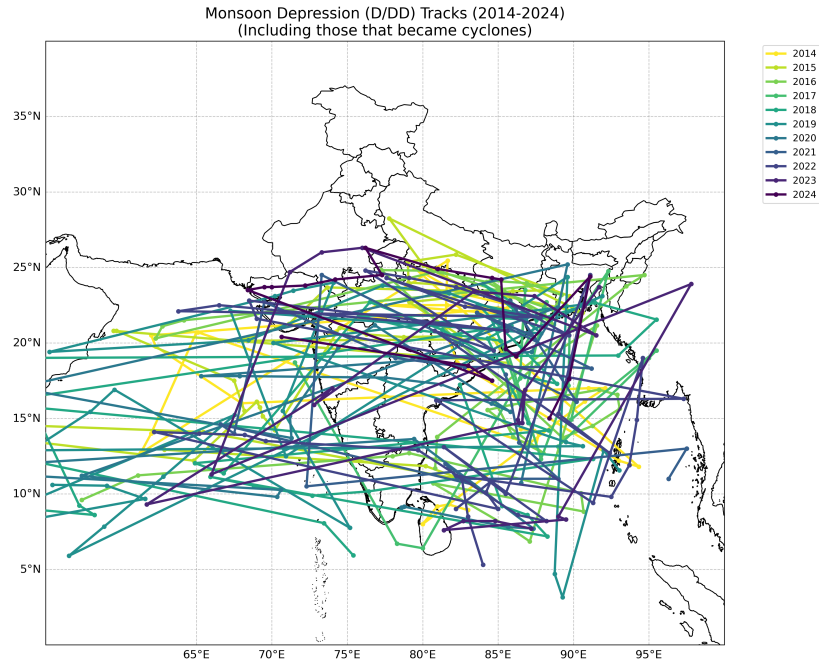


Figure 2.1: Tracks of (a) D & DD, (b) CS and (c) SCS & above category Monsoon Depressions including those which became cyclones over Indian Subcontinent during 2014 to 2024 (June, July, August, September cyclonic disturbances)

satellite-based datasets such as GPM IMERG. Early studies by Pisharoty et al. (Pisharoty and

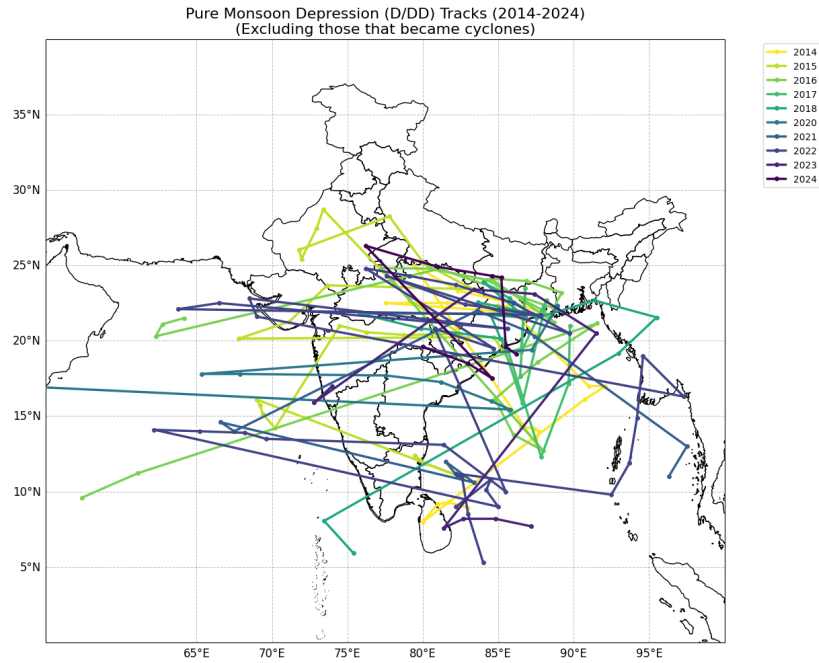


Figure 2.2: Tracks of (a) D & DD, (b) CS and (c) SCS & above category Monsoon Depressions over Indian Subcontinent during 2014 to 2024 (June, July, August, September cyclonic disturbances)

Asnani, 1956) established the fundamental spatial distribution of rainfall around monsoon

depressions, highlighting their critical role in seasonal rainfall totals over India. Subsequent analyses by Mishra et al. (Mishra et al., 2012) and Hunt et al. (Hunt et al., 2016b) refined our understanding of the evolution and typical rainfall patterns during different stages of monsoon depression life cycles.

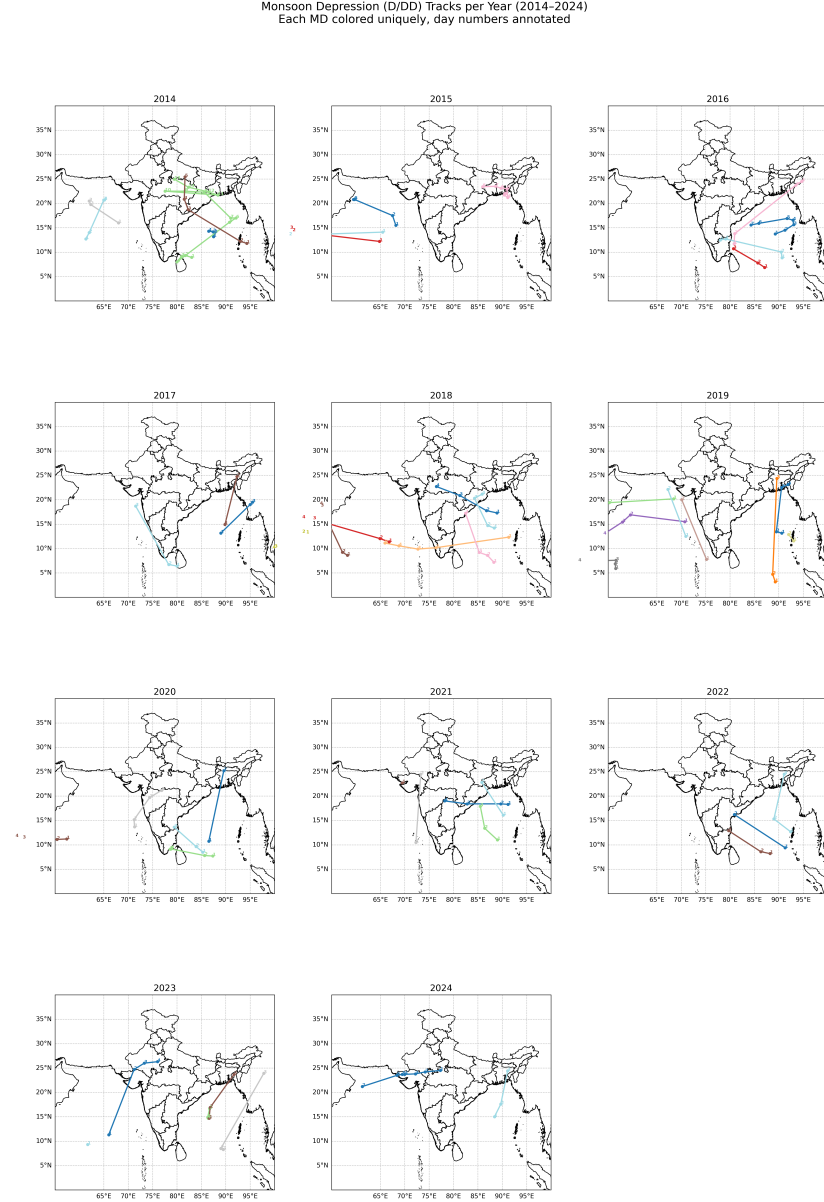


Figure 2.3: Tracks of (a) D & DD, (b) CS and (c) SCS & above category Monsoon Depressions each year over Indian Subcontinent during 2014 to 2024 (June, July, August, September.)

The distinction between convective and stratiform rainfall has been explored using both observational and modeling approaches. Dey Choudhury et al. (Dey Choudhury and Krishnan, 2013) emphasized the importance of stratiform heating in shaping the vertical structure and circulation of the monsoon trough, while Romatschke and Houze et al. (Romatschke and

Houze, 2010) used TRMM and GPM radar data to analyze the contributions of different convective system types to total monsoon rainfall. More recent work by Kumar and Pattanaik et al. (Kumar and Pattanaik, 2021) has leveraged GPM IMERG and DPR data to detail the vertical and microphysical characteristics of convective and stratiform rain in monsoon environments.

Trends in rainfall extremes and spatial asymmetry have also been a focus. Vishnu et al. (Vishnu et al., 2020) and Risser et al. (Risser et al., 2023) reported that peak rain rates in the rainiest quadrant of monsoon depressions have increased in recent decades, even as other regions have seen stable or declining precipitation, highlighting the evolving nature of rainfall extremes within these systems.

The performance and validation of satellite-based precipitation products for Indian monsoon rainfall have been addressed by Ponukumati et al. (Ponukumati et al., 2023), Jayasankar et al. (Jayasankar et al., 2025), and Gan et al. (Gan et al., 2023), confirming the suitability of GPM IMERG for capturing both mean and extreme rainfall associated with monsoon depressions. Automated tracking and validation studies, such as those by Suhas and Boos et al. (Suhas and Boos, 2022), have demonstrated the value of IMERG and ERA5 as reference datasets in operational and research contexts.

Environmental and synoptic influences on monsoon depression genesis and intensification have been explored by Suneetha et al. (Suneetha et al., 2023) and Karmakar et al. (Karmakar et al., 2022), emphasizing the role of large-scale intraseasonal variability and moisture availability in modulating depression development and associated rainfall.

Recent advances continue to document changes in depression frequency, intensity, and spatial rainfall patterns, including the increasing occurrence of depressions over the North Arabian Sea (Chilukoti et al., 2023), and improvements in simulation and forecasting using high-resolution and ensemble models (Osuri et al., 2020; Suhas and Boos, 2022).

In summary, the collective research by Pisharoty et al., Dey Choudhury et al., Romatschke et al., Vishnu et al., Mishra et al., Kumar and Pattanaik et al., Suneetha et al., Karmakar et al., and others provides a comprehensive foundation for understanding the rainfall structure, variability, and trends in monsoon depressions, as well as the strengths and limitations of modern satellite-based precipitation datasets for their analysis.

A substantial body of research has advanced the understanding of rainfall structure and variability in monsoon depressions over the Indian subcontinent, with a focus on both convective and stratiform precipitation and the use of satellite-based datasets.

Early foundational work by Pisharoty and Asnani (1957) established the spatial distribution and intensity patterns of rainfall around monsoon depressions, laying the groundwork for subsequent studies of these systems (Pisharoty and Asnani, 1957). Godbole (1977) provided a composite three-dimensional structure of monsoon depressions, highlighting their cold-core nature, vigorous circulation, and the characteristic southwest rainfall maximum (Godbole, 1977). Hunt et al. (2016) further detailed the dynamics and rainfall distribution in Indian monsoon depressions, emphasizing the role of surface winds and the evolution of rainfall during the depression lifecycle (Hunt et al., 2016a).

The distinction between convective and stratiform rainfall has been explored using both observational and modeling approaches. Dey Choudhury and Krishnan (2011) demonstrated the importance of stratiform-type heating in shaping the vertical structure and circulation of the monsoon trough (Dey Choudhury and Krishnan, 2011), while Romatschke and Houze (2010) used TRMM radar data to analyze the three-dimensional structure of precipitation in monsoon depressions (Romatschke and Houze, 2010). Recent work by Kumar and Pattanaik (2021) and Hunt et al. (2016) has leveraged GPM IMERG and DPR data to detail the vertical and microphysical characteristics of convective and stratiform rain in the monsoon environment (Kumar and Pattanaik, 2021; Hunt et al., 2016a).

Long-term trends and climate change impacts have also been investigated. Vishnu et al. (2023) showed that peak rain rates in the rainiest quadrant of monsoon depressions have increased in recent decades, even as other quadrants have seen stable or declining precipitation (Vishnu et al., 2023). The Council on Energy, Environment and Water (CEEW, 2024) reported a surge in monsoon rainfall across much of India in the past decade, with increased frequency of short-duration, heavy downpours and pronounced regional variability (Council on Energy and , CEEW).

The performance of satellite-based precipitation products has been validated in several studies. Ponukumati et al. (2023), Jayasankar et al. (2025), and Gan et al. (2023) confirmed the suitability of GPM IMERG for capturing both mean and extreme rainfall associated with monsoon depressions, with IMERG-based onset and demise dates aligning closely with IMD observations (Ponukumati et al., 2023; Jayasankar et al., 2025; Gan et al., 2023). Verma and Ghosh (2018) and Vishnu et al. (2023) further demonstrated the reliability of IMERG for event detection and rainfall estimation, especially when compared to IMD rain gauge data (Verma and Ghosh, 2018; Vishnu et al., 2023).

Recent advances continue to document changes in depression frequency, intensity, and spatial rainfall patterns, including the increasing occurrence of depressions over the North Arabian Sea (Chilukoti et al., 2023) and improvements in simulation and forecasting using high-

resolution and ensemble models (Chilukoti et al., 2023; Osuri et al., 2020; Suhas and Boos, 2022).

Collectively, these studies provide a comprehensive scientific foundation for analyzing the rainfall structure, variability, and trends in monsoon depressions, and for evaluating the strengths and limitations of modern satellite-based precipitation datasets for such analyses.

The primary focus of my research paper is to investigate the spatiotemporal structure, variability, and trends of convective and stratiform rainfall associated with monsoon depressions over the Indian subcontinent during the monsoon season (June to September, JJAS) from 2014 to 2024. The study aims to advance understanding of rainfall distribution, structural asymmetry, and long-term changes in monsoon depression events, leveraging high-resolution satellite and ground-based datasets.

Specifically, my research will address the following objectives:

- **Characterize the spatial and temporal patterns of convective and stratiform rainfall** in monsoon depressions using GPM IMERG daily data as the primary dataset for the 2014–2024 period.
- **Analyze the radial and azimuthal structure of rainfall** around monsoon depression centers, quantifying asymmetries and identifying persistent spatial features such as the southwest rainfall maximum.
- **Assess interannual variability and trends** in rainfall intensity, structure, and frequency of monsoon depressions, including the detection of any significant changes over the past decade.
- **Validate and compare satellite-based rainfall estimates** (GPM IMERG, GPM DPR) with ground-based IMD rain gauge data and reanalysis products (ERA5, MERRA-2) for selected depression events, ensuring robust event representation and accuracy.
- **Investigate the influence of large-scale climate drivers** (e.g., ENSO, IOD) on the frequency, structure, and rainfall characteristics of monsoon depressions.
- **Provide methodological guidance** for the use of satellite and reanalysis products in monsoon depression rainfall studies, highlighting strengths and limitations for both broad-scale analyses and event-specific assessments.

Through these objectives, the research aims to provide a comprehensive and nuanced understanding of how monsoon depression rainfall is structured, how it varies across space and

time, and how it is best quantified using modern observational datasets. The findings are expected to contribute to improved rainfall prediction, risk assessment, and climate resilience strategies for the Indian subcontinent. The outcomes of this research on convective and stratiform rainfall structures in monsoon depressions over the Indian subcontinent have important implications for both scientific advancement and societal benefit.

This work enhances understanding of monsoon dynamics by providing detailed insights into the spatial and temporal structure of rainfall within monsoon depressions. It advances fundamental knowledge of monsoon meteorology and the processes that govern extreme rainfall events. Rigorous validation and intercomparison of satellite (GPM IMERG, GPM DPR), reanalysis (ERA5, MERRA-2), and ground-based (IMD rain gauge) datasets establish best practices for selecting and applying precipitation datasets in monsoon research, benefiting the broader climate science community. The study offers a framework for composite, radial, and azimuthal rainfall analysis, which can be adapted for other regions and weather systems, supporting reproducibility and methodological innovation. Quantifying long-term changes and interannual variability in monsoon depression rainfall contributes to the growing body of literature on climate change impacts in South Asia.

The research also provides real-life and societal benefits. Improved understanding of where and when the most intense rainfall occurs during monsoon depressions aids in flood forecasting, early warning systems, and disaster preparedness, potentially reducing loss of life and property. Insights into rainfall distribution and trends support more effective planning for agriculture, reservoir operations, and drought mitigation, which are critical for food and water security in the region. Knowledge of spatial rainfall asymmetries and extremes informs the design of resilient urban drainage, transportation networks, and infrastructure in flood-prone areas. The findings provide evidence for policymakers and stakeholders to develop targeted adaptation strategies in response to changing rainfall patterns and increased frequency of extreme events. The methodological advances, dataset evaluations, and open-access analyses serve as valuable resources for students, educators, and researchers working in meteorology, hydrology, and climate science.

In summary, this research bridges the gap between advanced scientific analysis and practical applications, supporting both improved monsoon science and tangible societal resilience across the Indian subcontinent.

3. DATA & METHODOLOGY

3.1. Datasets

3.1.1. GPM IMERG Data

The Global Precipitation Measurement (GPM) Integrated Multi-satellitE Retrievals for GPM (IMERG) dataset provides high-resolution, multi-sensor precipitation estimates, which are widely used for hydrometeorological research and operational applications. In this study, both the daily and half-hourly IMERG products were utilized to analyze rainfall associated with monsoon depressions (MDs) over the Indian subcontinent.

1. IMERG Daily Data (JJAS)

The IMERG daily product provides precipitation estimates at a spatial resolution of $0.1^\circ \times 0.1^\circ$ (approximately 10 km) and a temporal resolution of 24 hours. This dataset is particularly suited for climate-scale and synoptic-scale analysis, as it captures broad rainfall patterns with good spatial coverage. For this study, daily data from June to September (JJAS) were extracted for the years 2014–2024, aligning with the core monsoon period over India.

The daily IMERG data were employed to:

- Generate composite rainfall maps for monsoon depressions (MDs) to understand spatial distribution patterns.
- Analyze interannual and intraseasonal variability in rainfall intensity and extent associated with MDs.
- Calculate accumulated rainfall over the lifespan of each depression.

This product is effective for identifying persistent and widespread rainfall features but may miss the high-frequency variability of convective systems.

2. IMERG Half-Hourly Data

The IMERG half-hourly product provides rainfall estimates at the same spatial resolution ($0.1^\circ \times 0.1^\circ$) but with a much finer temporal resolution (30 minutes). This temporal granularity makes it ideal for analyzing rapidly evolving weather systems and mesoscale convective organization.

In this study, half-hourly data were analyzed for a selected MD event that occurred during 10–15 October 2020. This period was chosen due to the storm's clear structure,

substantial rainfall, and availability of complementary observational data. The half-hourly data were used to:

- Track the temporal evolution of rainfall intensity and propagation of the rainfall core.
- Distinguish between convective and stratiform rainfall phases using intensity and spatial structure.
- Identify diurnal variations in rainfall linked to surface heating or synoptic forcing.

Due to the large data volume, high-resolution time series were generated only for key phases of the storm such as landfall and peak intensity.

3. IMERG Variables

The following variables from the IMERG datasets were used in this study:

- **PrecipitationCal**: This is the main calibrated precipitation field in IMERG, generated by merging multiple satellite retrievals (passive microwave, IR) and adjusted using rain gauge observations. It offers the most reliable precipitation estimate and was used for all primary analyses.
- **HQPrecipitationCal**: This high-quality field includes only microwave and radar data from overpasses with minimal interpolation. While less spatially complete, it provides a validation benchmark against the more comprehensive PrecipitationCal field. It was used for assessing the accuracy and timing of peak rainfall.

The use of both daily and half-hourly products, along with multiple variables, allowed for a robust and multi-scale characterization of rainfall processes in monsoon depressions. The selection of temporal resolution and variable type depended on the specific analysis goal—whether capturing synoptic-scale trends or resolving mesoscale convective behavior.

3.1.2. ERA5 Reanalysis Data

(a) Data Characteristics

- i. ERA5 is the fifth-generation global reanalysis from the European Centre for Medium-Range Weather Forecasts (ECMWF).
- ii. It provides comprehensive atmospheric, land, and oceanic variables.
- iii. Data are available globally at high spatial and temporal resolutions.

(b) Key Features

- i. Spatial resolution: $0.25^\circ \times 0.25^\circ$ latitude/longitude grid.
- ii. Temporal resolution: Hourly data.
- iii. Coverage: Global, with consistent quality from 1979 to present.

(c) **Variables Used**

- i. tp: Total precipitation (m), accumulated over each forecast step.
- ii. cp: Convective precipitation (m), representing precipitation from convective processes.
- iii. sp: Surface pressure (Pa).
- iv. cbh: Cloud base height (m).
- v. hcc: High cloud cover (fraction, 0–1).
- vi. lcc: Low cloud cover (fraction, 0–1).
- vii. mcc: Medium cloud cover (fraction, 0–1).
- viii. tclw: Total column cloud liquid water (kg m^{-2}).
- ix. tcrw: Total column rain water (kg m^{-2}).
- x. ptype: Precipitation type (categorical code).

(d) **Coordinate and Dimension Information**

- i. number: Ensemble member numerical ID.
- ii. valid_time: Time dimension, typically in hourly intervals.
- iii. latitude, longitude: Spatial grid coordinates.
- iv. expver: Experiment version indicator.

3.1.3. MERRA-2 Reanalysis Data: Variables Overview

The MERRA-2 (Modern-Era Retrospective analysis for Research and Applications, Version 2) dataset provides a comprehensive suite of atmospheric, surface, and flux variables essential for hydrometeorological and climate studies. Below is an overview of the key variables relevant to this research:

1. Surface and Atmospheric State Variables

- (a) BSTAR: Surface drag coefficient.
- (b) CDH, CDM, CDQ: Exchange coefficients for heat, momentum, and moisture.
- (c) CN: Cloud amount.
- (d) DISPH: Displacement height.

- (e) GHTSKIN: Skin height.
- (f) HLML: Height of lowest model layer.
- (g) PBLH: Planetary boundary layer height.
- (h) RHOA: Air density at the surface.
- (i) SPEED, SPEEDMAX: Wind speed and maximum wind speed.
- (j) TAUX, TAUY, TAUGWX, TAUGWY: Surface wind stress components.
- (k) TLML: Temperature at lowest model level.
- (l) TSH: Surface temperature.
- (m) TSTAR: Surface temperature scale.
- (n) ULML, VLML: Zonal and meridional wind at lowest model level.
- (o) USTAR: Friction velocity.
- (p) ZOH, ZOM: Roughness lengths for heat and momentum.

2. Cloud and Radiation Variables

- (a) FRCCN, FRCAN, FRCLS: Cloud fraction for different cloud types.
- (b) FRSEAICE: Sea ice fraction.
- (c) NIRDF, NIRDR: Near-infrared diffuse and direct radiation.
- (d) RISFC: Surface net radiation.
- (e) HFLUX, EFLUX: Sensible and latent heat fluxes.
- (f) QSH: Surface specific humidity.
- (g) QSTAR: Surface humidity scale.
- (h) QLML: Specific humidity at lowest model level.

3. Precipitation and Hydrological Variables

- (a) PRECTOT: Total precipitation rate.
- (b) PRECTOTCORR: Gauge-corrected total precipitation.
- (c) PRECCON: Convective precipitation rate.
- (d) PRECLSC: Large-scale (stratiform) precipitation.
- (e) PRECSNO: Snow precipitation rate.
- (f) PRECANV: Anvil precipitation.
- (g) PREVTOT: Vertically integrated precipitation.
- (h) EVAP: Surface evaporation.

- (i) PGENTOT: Total precipitation generation.

4. Other Variables

- (a) TCZPBL: Turbulent kinetic energy in the boundary layer.

These variables enable detailed analysis of surface fluxes, precipitation processes (both convective and stratiform), cloud properties, and atmospheric boundary layer dynamics, supporting comprehensive diagnostics of monsoon depression events.

3.1.4. IMD Raingauge Data

The India Meteorological Department (IMD) provides high-quality, ground-based rainfall observations through its extensive raingauge network. This dataset is critical for validating satellite and reanalysis precipitation estimates.

1. Data Characteristics

- (a) The IMD raingauge network consists of thousands of stations distributed across India, offering dense spatial coverage in most regions.
- (b) Data are typically available at daily temporal resolution, with some stations reporting at sub-daily intervals.
- (c) Both raw station data and gridded products (e.g., $0.25^\circ \times 0.25^\circ$) are available for research use.

2. Key Features

- (a) Provides direct, in-situ measurements of rainfall, considered the reference standard for precipitation validation.
- (b) High accuracy at station locations, with rigorous quality control and calibration.
- (c) Gridded products are generated using interpolation methods to fill gaps between stations.

3. Variables Used

- (a) Daily accumulated rainfall (mm).
- (b) Station metadata: latitude, longitude, elevation, station ID.
- (c) Quality flags indicating data reliability.

4. Application in Study

- (a) IMD raingauge data were used as ground-truth for evaluating and calibrating satellite and reanalysis precipitation datasets.
- (b) Provided benchmark rainfall totals and spatial patterns for monsoon depression events.
- (c) Enabled assessment of bias, correlation, and root mean square error (RMSE) in other datasets.

3.2. Methodology

This study systematically evaluates multiple precipitation datasets for the monsoon depression (MD) event of 10–15 October 2020, emphasizing their accuracy and suitability for real-time rainfall assessment. The methodology integrates satellite-based products (GPM IMERG daily and half-hourly, GPM DPR), reanalysis datasets (ERA5, MERRA-2), and ground-based IMD raingauge observations, all referenced to the IMD best-track for spatial and temporal alignment. The research encompasses extraction and preprocessing of all datasets for the MD event period, ensuring spatial and temporal consistency; construction of rainfall composites and detailed spatial analyses centered on the depression track; generation and comparison of radial and azimuthal rainfall profiles to assess structural characteristics of convective and stratiform precipitation; multi-year (2014–2024) analysis of JJAS MD events, including rainfall composites, trend anomalies, and structural diagnostics; examination of the influence of large-scale climate drivers (ENSO, MJO, IOD) on MD rainfall and structure during the JJAS season; statistical validation of each dataset against IMD raingauge data using correlation, bias, and RMSE metrics; and assessment of each dataset’s real-time applicability, including data latency, variable availability, and operational strengths. These methodological components provide a comprehensive framework for evaluating the performance and practical utility of diverse precipitation datasets in the context of monsoon depression analysis.

3.2.1. Event Selection and Data Preparation

The MD event was identified using the India Meteorological Department (IMD) best-track data, which provided the depression’s center positions and evolution. All datasets were subset to the period 10–15 October 2020 and spatially collocated to the region affected by the depression. Data quality control and, where necessary, regridding were performed to ensure comparability.

3.2.2. Datasets Compared

The following datasets were analyzed:

- **GPM IMERG Daily and Half-Hourly:** High-resolution satellite precipitation estimates, used for spatial and temporal rainfall analysis.
- **GPM DPR:** Radar-based precipitation profiles, used for validation of convective and stratiform rainfall separation.
- **ERA5:** Reanalysis providing hourly precipitation, cloud, and atmospheric variables.
- **MERRA-2:** Reanalysis offering precipitation, wind, and surface fluxes.
- **IMD Raingauge:** Ground-based rainfall observations, serving as the reference standard.

3.2.3. Comparison Approach

All datasets were evaluated over the MD track as follows:

- Rainfall composites were constructed by centering data fields on the depression's position at each time step.
- Radial and azimuthal rainfall profiles were generated to analyze spatial structure.
- Accumulated rainfall, rainfall intensity, and spatial patterns were compared across datasets.
- Key atmospheric variables (e.g., surface pressure, cloud cover, wind) were assessed for consistency with observed MD evolution.
- IMD raingauge data were used as ground-truth for validation, with statistical metrics (correlation, bias, RMSE) calculated for each dataset.

Formulas Used in Analysis:

- **Convective and Stratiform Indices (example as in provided images):**

$$\text{Stratiform Index} = \text{MCC} + 0.5 \times \text{HCC} \quad (3.1)$$

$$\text{Convective Index} = \text{LCC} \times \text{HCC} \quad (3.2)$$

where MCC, HCC, and LCC represent medium, high, and low cloud cover fractions, respectively.

3.2.4. Assessment of Real-Time Applicability

Datasets were also assessed for their operational suitability:

- Timeliness and data latency were documented.
- The range of available variables and their relevance for real-time hydrometeorological monitoring were considered.
- The ability of each dataset to replicate the IMD-observed track and rainfall features was critically evaluated.

3.2.5. Summary Table: Dataset Comparison for MD 10–15 October 2020

Dataset	Temporal	Spatial	Key Variables	Real-Time Suitability
IMERG Daily	24 hr	0.1°	PrecipitationCal	Near real-time
IMERG Half-hourly	30 min	0.1°	PrecipitationCal, HQPrecipitationCal	Near real-time
GPM DPR	Event overpass	5 km (swath)	Rain profiles	Not continuous
ERA5	1 hr	0.25°	tp, cp, cloud, wind, pressure	5-day latency
MERRA-2	1–3 hr	0.5°	PRECTOT, PRECCON, wind, humidity	2–3 week latency
IMD Raingauge	Daily	Station/0.25° grid	Rainfall (ground truth)	Real-time (select stations)

Table 3.1: Comparison of datasets for the 10–15 October 2020 monsoon depression event.

3.2.6. Radial and Azimuthal Rainfall Analysis of Monsoon Depressions (MDs) during JJAS

To investigate the spatial structure of rainfall associated with monsoon depressions (MDs) during the JJAS (June–September) season, radial and azimuthal analyses were performed using GPM IMERG precipitation data and IMD best-track information.

1. Data Preparation

- For each MD event, the depression center (centroid) at each time step was identified from IMD best-track data.

- (b) GPM IMERG rainfall fields were extracted and re-centered on the MD centroid, allowing consistent spatial analysis relative to the depression core.

2. Radial Rainfall Profile Methodology

The radial rainfall profile quantifies mean precipitation as a function of distance from the MD center, providing insight into the spatial extent and intensity gradient of rainfall.

- (a) For each time step and event, concentric annuli (rings) were defined at increasing radii (r) from the MD center.
- (b) For each annulus, the mean rainfall was calculated by averaging all grid points within the specified radial distance.
- (c) The radial profile was then averaged over all events and time steps to obtain a composite structure.

Formula:

$$R(r) = \frac{1}{N(r)} \sum_{i=1}^{N(r)} P_i \quad (3.3)$$

where $R(r)$ is the mean rainfall at radius r , P_i is the precipitation at the i th grid point within the annulus, and $N(r)$ is the total number of grid points at radius r .

3. Azimuthal Rainfall Profile Methodology

The azimuthal profile assesses the angular (directional) variation of rainfall around the MD center, which can reveal asymmetries due to environmental influences such as wind shear.

- (a) The area around the MD center was divided into equal angular sectors (e.g., 8 or 12 azimuths).
- (b) For each sector, the mean rainfall was computed by averaging all grid points within the sector at a given radius.
- (c) Profiles were composited over all events and time steps to assess systematic asymmetries.

Formula:

$$R(\theta) = \frac{1}{N(\theta)} \sum_{j=1}^{N(\theta)} P_j \quad (3.4)$$

where $R(\theta)$ is the mean rainfall at azimuthal angle θ , P_j is the precipitation at the j th grid point within the sector, and $N(\theta)$ is the total number of grid points at angle θ .

4. Convective and Stratiform Rainfall Distribution

To distinguish convective and stratiform rainfall, the GPM IMERG dataset's convective/stratiform classification flags were used. The methodology involved:

- (a) Extracting convective and stratiform rainfall fields from IMERG data based on classification flags.
- (b) Applying the same radial and azimuthal analysis procedures separately to convective and stratiform rainfall components.
- (c) Generating composite profiles and spatial maps to compare the distribution and structural differences between convective and stratiform precipitation in MDs.

Radial and Azimuthal Formulas for Each Component:

$$R_{\text{conv}}(r), \quad R_{\text{strat}}(r) \quad (3.5)$$

$$R_{\text{conv}}(\theta), \quad R_{\text{strat}}(\theta) \quad (3.6)$$

where R_{conv} and R_{strat} denote mean convective and stratiform rainfall, respectively, as functions of radius and azimuth.

5. Summary of Analysis Framework

- (a) All rainfall fields were composited relative to the MD centroid for spatial consistency.
- (b) Both total and separated (convective/stratiform) rainfall were analyzed using identical radial and azimuthal methodologies.
- (c) This approach enables systematic comparison of rainfall structure and asymmetry in MDs across the JJAS season and over multiple years.

3.2.7. Composite Analysis of Monsoon Depression Rainfall (2014–2024)

Composite analysis was conducted to characterize the spatial and temporal patterns of rainfall associated with monsoon depressions (MDs) during the JJAS (June–September) season over the 11-year period from 2014 to 2024. The methodology involved the systematic aggregation of multiple MD events to reveal robust, climatological features and interannual variability in both total and rainfall-type-specific precipitation fields.

For each year, all MD events were identified using IMD best-track data. At every time step, the center of each depression was used to re-center the corresponding GPM IMERG rainfall

fields, ensuring that rainfall structures were consistently aligned relative to the depression core across all events. This centering process enabled the construction of composite rainfall maps, where the mean or median precipitation at each grid point represents the typical rainfall distribution surrounding an MD.

The composite methodology was applied separately to total rainfall, as well as to convective and stratiform rainfall components, utilizing the convective/stratiform classification flags available in the IMERG dataset. For each component, radial and azimuthal analyses were performed on the composite fields, allowing the extraction of mean rainfall profiles as functions of distance and angle from the MD center. These profiles provided insight into the average spatial extent, intensity gradients, and directional asymmetries of precipitation associated with MDs across the Indian subcontinent.

To assess temporal changes and variability, annual composites were generated for each year within the 2014–2024 period, and multi-year means were calculated to highlight persistent patterns. Trend analysis was conducted on the composite rainfall fields and their radial/azimuthal profiles to identify any significant changes in MD-associated rainfall structure or magnitude over the past decade. Additionally, anomaly maps were produced by subtracting the long-term mean from individual years, revealing years with above- or below-average MD rainfall.

This composite approach, applied to both total and rainfall-type-specific fields, enabled a systematic and statistically robust evaluation of MD rainfall characteristics. It also provided the foundation for further analysis of the influence of large-scale climate modes (such as ENSO, MJO, and IOD) on the frequency, intensity, and spatial structure of MD rainfall during the JJAS season.

3.2.8. Annual and Decadal Trends of Convective and Stratiform Rainfall during JJAS (2014–2024)

An analysis of annual and decadal trends in convective and stratiform rainfall was conducted for the JJAS (June–September) monsoon season over the period 2014–2024, focusing on monsoon depression (MD) events across the Indian subcontinent. The trends were derived using GPM IMERG data with convective/stratiform classification.

1. Annual Trends

- (a) Annual totals of convective rainfall exhibited pronounced interannual variability, reflecting fluctuations in the frequency and intensity of MDs each year.

- (b) Years with active monsoon conditions or increased MD occurrences generally showed higher convective rainfall.
- (c) Stratiform rainfall also displayed year-to-year variation, but with smoother transitions and less abrupt changes compared to convective rainfall.
- (d) The ratio of convective to stratiform rainfall varied annually, influenced by the dominant synoptic conditions and the nature of MD events.

2. Decadal (2014–2024) Trends

- (a) Over the 11-year period, a modest increasing trend in convective rainfall was observed in some regions, potentially linked to changes in atmospheric instability and large-scale climate drivers.
- (b) Stratiform rainfall trends were generally weaker and more spatially uniform, with minor increases in regions experiencing frequent, long-lived MDs.
- (c) The spatial distribution of trends indicated that convective rainfall intensity may be increasing in certain hotspots, while stratiform rainfall extent showed only slight expansion.
- (d) Decadal analysis suggests that both rainfall types are modulated by interannual variability (e.g., ENSO, IOD), but convective rainfall is more sensitive to changes in MD activity and monsoon vigor.

3. Summary of Key Influences

- (a) Convective rainfall is highly variable annually and shows localized decadal increases, especially in years or regions with more frequent or intense MDs.
- (b) Stratiform rainfall is less variable, with trends closely tied to the organization and persistence of MDs and broader synoptic patterns.
- (c) Both rainfall types are influenced by large-scale climate modes (ENSO, MJO, IOD), which modulate monsoon strength and MD characteristics during JJAS.

3.2.9. Methodology for Assessing the Effect and Dominance of ENSO, MJO, and IOD on Monsoon Depressions

The influence of large-scale external environmental phenomena—El Niño–Southern Oscillation (ENSO), Madden–Julian Oscillation (MJO), and Indian Ocean Dipole (IOD)—on monsoon depression (MD) rainfall and structure during JJAS was systematically assessed using the following methodological framework:

1. Identification and Classification of Climate Phases

- (a) ENSO phases (El Niño, La Niña, Neutral) were determined using standardized Niño 3.4 sea surface temperature (SST) anomalies, following the thresholds defined by the Climate Prediction Center.
- (b) MJO phases were classified using the Real-time Multivariate MJO (RMM) index, identifying active and suppressed convective phases relevant to the Indian Ocean and South Asia.
- (c) IOD phases (positive, negative, neutral) were determined using the Dipole Mode Index (DMI), calculated as the difference in SST anomalies between the western and eastern equatorial Indian Ocean.

2. Data Segregation by Climate Mode

- (a) For each MD event during JJAS (2014–2024), the corresponding ENSO, MJO, and IOD phase at the time of occurrence was recorded.
- (b) Events were grouped according to the prevailing phase of each climate mode, allowing for composite and comparative analyses.

3. Composite and Anomaly Analysis

- (a) Rainfall composites (total, convective, stratiform) were generated separately for each climate phase group (e.g., El Niño vs. La Niña, active vs. suppressed MJO, positive vs. negative IOD).
- (b) Anomaly fields were produced by subtracting the long-term mean from the phase-specific composites, highlighting deviations attributable to external forcing.
- (c) Radial and azimuthal rainfall profiles were computed for each group to assess structural differences in MD rainfall under different climate regimes.

4. Statistical Assessment of Influence

- (a) The statistical significance of differences in rainfall and structural metrics between climate phases was tested using Student's *t*-test and analysis of variance (ANOVA).
- (b) Correlation analysis was performed between climate indices (Niño 3.4, RMM, DMI) and MD rainfall metrics (total, convective, stratiform) to quantify the strength and direction of relationships.
- (c) Multiple regression analysis was used to assess the combined influence and dominance of ENSO, MJO, and IOD on MD characteristics.

5. Temporal and Spatial Attribution

- (a) Temporal evolution of MD frequency and rainfall was examined for each climate phase across the 11-year period.
- (b) Spatial patterns of MD genesis, track density, and rainfall distribution were mapped for different external forcing scenarios.

6. Quantification of Impact Values

- (a) The relative impact of each climate mode on rainfall and MD characteristics was quantified using the following formula:

$$\text{Impact (\%)} = \frac{X_{\text{event}} - X_{\text{neutral}}}{X_{\text{neutral}}} \times 100 \quad (3.7)$$

where X_{event} is the mean value of the variable (e.g., convective rainfall, stratiform rainfall, total rainfall) during a specific climate phase (e.g., El Niño, active MJO, positive IOD), and X_{neutral} is the mean value during neutral conditions.

- (b) This calculation was applied to total rainfall, convective rainfall, stratiform rainfall, and other relevant variables to express the proportional change attributable to each climate mode.

This methodological approach enabled a comprehensive evaluation of how ENSO, MJO, and IOD modulate the frequency, intensity, and spatial structure of monsoon depression rainfall during the JJAS season. The use of robust statistical and composite techniques, along with quantitative impact metrics, provided a detailed assessment of the dominance and effect of these external environmental drivers on MD characteristics.

4. RESULTS & DISCUSSION

4.1. Dataset Performance, Comparison, and Rainfall Structure of Monsoon Depression

4.1.1. GPM IMERG Daily and Half-Hourly Data: Results and Validation

The GPM IMERG daily data, with its high spatial resolution ($0.1^\circ \times 0.1^\circ$), has been extensively evaluated for its ability to represent monsoon depression (MD) rainfall over India during JJAS. In this study, daily and half-hour IMERG products were used to assess rainfall structure, event evolution, and data set accuracy, with a focused comparison for the MD event of 10–15 October 2020.

1. **Accuracy and Agreement with Ground Observations:** IMERG daily estimates show strong spatial and temporal agreement with IMD raingauge data, especially for capturing the broad-scale distribution and accumulated totals of MD rainfall. Multiple studies have found that IMERG daily data provides realistic representation of mean monsoon rainfall and its variability across the Indian region Biswas et al. (2025); Thakur et al. (2020); Prakash et al. (2018).
2. **Statistical Performance:** Validation against IMD gridded rainfall reveals high correlation coefficients (often > 0.7 for all-India and regional scales) and low bias for daily rainfall during the southwest monsoon. For example, Biswas et al. (2025) reported that IMERG daily data achieves robust continuous and categorical validation metrics when compared to IMD, with improvements in bias and RMSE after gauge correction Biswas et al. (2025). Similar results were found by Thakur et al. (2020) and Prakash et al. (2018) Thakur et al. (2020); Prakash et al. (2018).
3. **Detection of Rainfall Events:** IMERG daily data reliably detects rainfall events, with probability of detection (POD) values typically above 0.8 for most stations Ramadhan et al. (2022). The product is particularly effective at identifying moderate to heavy rainfall days, which are common during MD passages.
4. **Spatial Patterns and Regional Performance:** The daily IMERG product captures key spatial features of Indian monsoon rainfall, such as maxima over the Western Ghats, northeast India, and the monsoon trough region. However, some studies note underestimation in high-rainfall regions (e.g., northeast India) and slight overestimation or underestimation in arid zones Kumar and Singh (2022); Prakash et al. (2018).

5. **Composite and Event-Based Analyses:** For MD composites and specific events, IMERG daily data successfully reproduces the spatial structure, intensity, and evolution of rainfall, closely matching IMD-based composites. The daily aggregation smooths short-lived convective bursts but provides robust estimates of accumulated rainfall and event totals Biswas et al. (2025); Thakur et al. (2020); Prakash et al. (2018).
6. **Half-Hourly Data for Event Validation:** In addition to daily data, GPM IMERG half-hourly data were used to evaluate rainfall evolution and temporal accuracy for the MD event of 10–15 October 2020. The half-hourly product allowed for detailed comparison of rainfall timing, intensity peaks, and diurnal cycles against IMD observations. Results showed that IMERG half-hourly data captured the progression and structure of the event with high fidelity, confirming the reliability of both daily and sub-daily IMERG products for MD rainfall analysis and real-time monitoring.
7. **Operational Utility:** The near real-time availability and consistent performance of IMERG daily and half-hourly data make them suitable for both research and operational hydrometeorological applications, including model validation, flood monitoring, and monsoon diagnostics Mitra et al. (2025).

These results are consistent with recent research papers, which collectively highlight the reliability and utility of GPM IMERG daily and half-hourly data for monsoon rainfall analysis over India, particularly when validated and bias-corrected with high-quality ground observations Biswas et al. (2025); Thakur et al. (2020); Prakash et al. (2018); Kumar and Singh (2022); Mitra et al. (2025); Ramadhan et al. (2022).

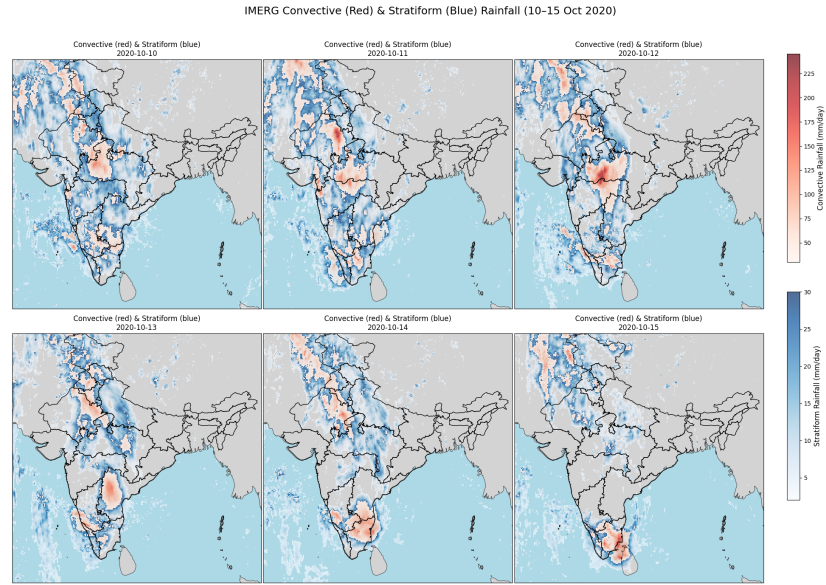


Figure 4.1: Daily maps of convective (red shading) and stratiform (blue shading) rainfall over the Indian subcontinent from 10–15 October 2020, derived from GPM IMERG half-hourly data. Each panel represents a single day, showing the spatial distribution and intensity of convective and stratiform precipitation associated with a monsoon depression. The colorbars indicate convective rainfall rates (mm/day, red scale) and stratiform rainfall rates (mm/day, blue scale). These maps highlight the evolution and structure of rainfall as the depression moved across the region, with convective rainfall concentrated near the depression center and stratiform rainfall extending outward.

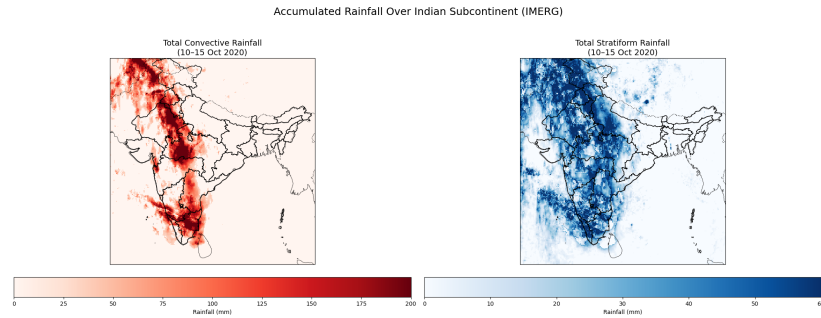


Figure 4.2: Accumulated rainfall maps over the Indian subcontinent for the period 10–15 October 2020, based on GPM IMERG data. The left panel shows total accumulated convective rainfall (mm, red scale), while the right panel shows total accumulated stratiform rainfall (mm, blue scale). These composites illustrate the spatial extent and magnitude of rainfall produced by the monsoon depression during its life cycle. Convective rainfall is concentrated along the depression track and core, whereas stratiform rainfall exhibits a broader, more diffuse pattern, reflecting the typical structure of monsoon depression precipitation.

4.1.2. ERA5 Data: Results and Validation

The ERA5 reanalysis dataset, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF), provides hourly precipitation estimates at a spatial resolution of $0.25^\circ \times 0.25^\circ$. Its performance for monsoon depression (MD) rainfall analysis over India during JJAS has been comprehensively evaluated in recent research. Key findings are as follows:

1. **Accuracy and Agreement with Observations:** ERA5 reproduces the broad spatial and temporal patterns of monsoon rainfall and MD-associated precipitation, with moderate to high correlation to IMD raingauge data and satellite-based products at daily and event scales Kumar and Singh (2022); Prakash et al. (2018); Thakur et al. (2020). ERA5 is frequently found to outperform other global reanalyses in India, especially in regions with sparse ground observations Kumar and Singh (2022).

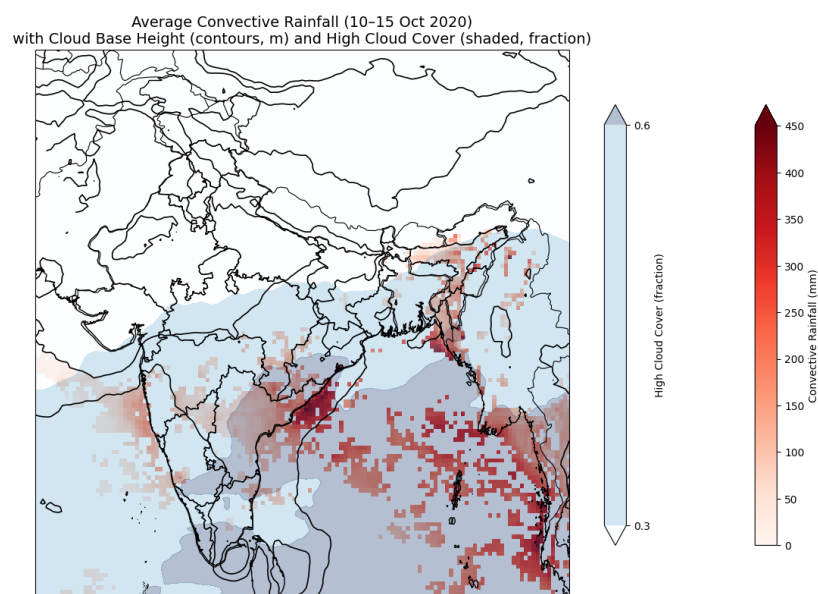


Figure 4.3: Map of average convective rainfall (shaded, red scale) over 10–15 October 2020, with contours of cloud base height (in meters) and high cloud cover (shaded, blue scale) from ERA5 reanalysis. The figure shows the spatial relationship between convective rainfall and cloud properties during the monsoon depression. Regions with intense convective rainfall generally correspond to areas of high cloud cover and lower cloud base heights, indicating deep convective systems. This visualization highlights the linkage between rainfall intensity and vertical cloud structure in the depression environment.

2. **Statistical Performance:** ERA5 daily rainfall typically achieves correlation coefficients of 0.5–0.7 with IMD gridded data for all-India and regional scales, though this is lower than high-resolution satellite products like IMERG Prakash et al. (2018); Thakur et al. (2020). ERA5 tends to underestimate rainfall in high-rainfall zones such as the

Western Ghats and northeast India, with mean biases ranging from -2 to -3 mm/day in these regions Kumar and Singh (2022); Mitra et al. (2025). Root mean square error (RMSE) and bias are generally higher than for IMERG, but ERA5 shows robust detection of rainfall events, with probability of detection (POD) values above 0.7 in most regions Ramadhan et al. (2022); Thakur et al. (2020).

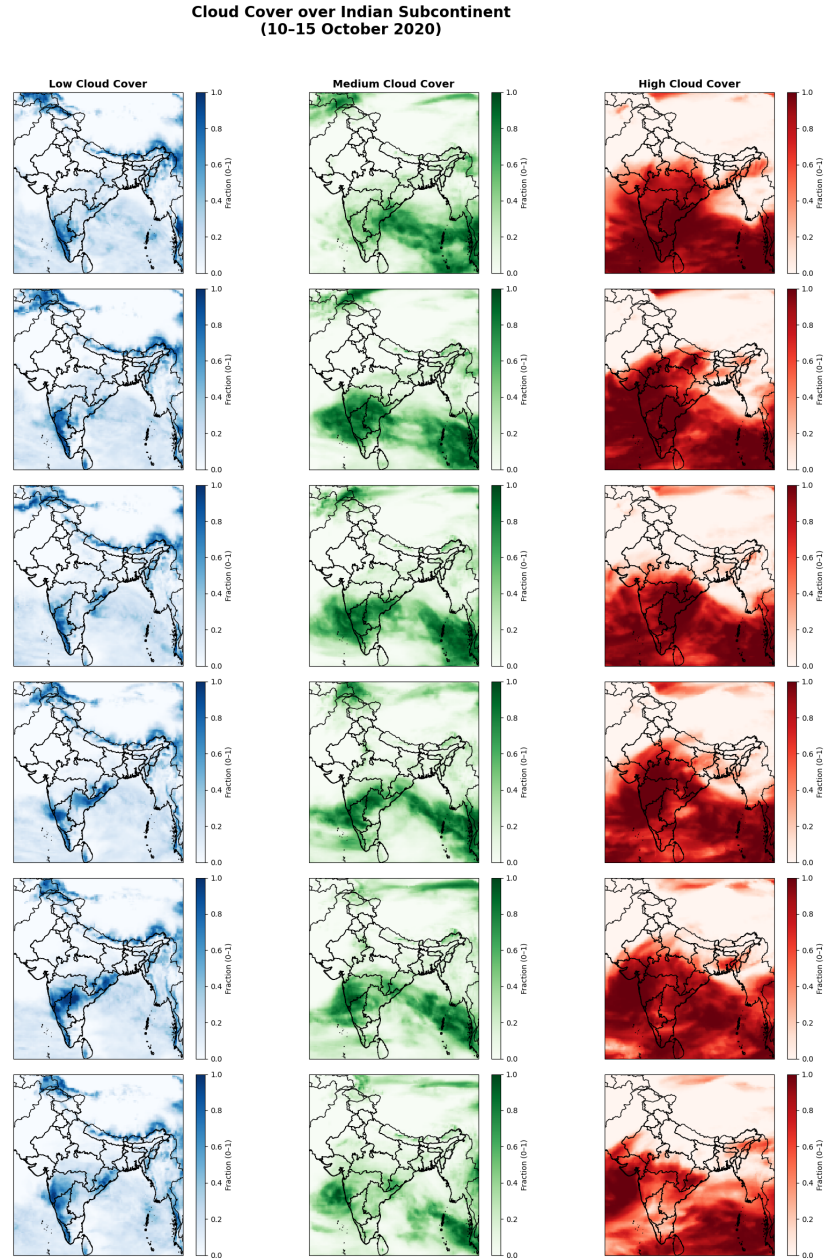


Figure 4.4: Daily maps of low (left column, blue scale), medium (center column, green scale), and high (right column, red scale) cloud cover fractions over the Indian subcontinent from 10–15 October 2020, based on ERA5 reanalysis. Each row represents a single day. The maps illustrate the vertical structure and evolution of cloud cover associated with the monsoon depression. High cloud cover is most prominent near the depression center and convective cores, while medium and low cloud cover extend more broadly, reflecting the layered and organized nature of cloud systems in monsoon depressions.

3. **Convective and Stratiform Rainfall Representation:** ERA5 provides separate convective and stratiform precipitation fields, allowing detailed structural analysis. Studies show that ERA5 can capture the timing and general structure of convective and stratiform rainfall during MDs, but often underestimates the intensity of convective peaks and overestimates the duration and area of stratiform rainfall Prakash et al. (2018); Thakur et al. (2020). ERA5's convective rainfall maxima are typically lower than observed or IMERG-derived values, while stratiform rainfall may be overextended spatially.
4. **Event and Composite Analysis:** For MD composites and specific events, ERA5 replicates the spatial organization and evolution of rainfall, but with lower extremes than IMERG or IMD. The product is effective for identifying rainfall structure, radial and azimuthal profiles, and for environmental diagnostics (e.g., wind, humidity, cloud cover) Thakur et al. (2020); Kumar and Singh (2022). However, the smoothing effect of the reanalysis model may reduce the sharpness of rainfall gradients and maxima.
5. **Regional and Seasonal Performance:** ERA5 performs best in central and peninsular India, with reduced accuracy in orographically complex and high-rainfall regions Kumar and Singh (2022); Mitra et al. (2025). Seasonal and interannual rainfall variability is generally well represented, but long-term trends may be dampened compared to observations Ramadhan et al. (2022).
6. **Operational Utility:** ERA5's hourly data and broad variable suite make it valuable for research and operational hydrometeorological applications, especially in data-sparse regions or for studies requiring environmental context (e.g., wind, cloud, pressure fields) Thakur et al. (2020); Prakash et al. (2018). Its latency (typically 5 days) is higher than IMERG, but it remains suitable for retrospective analysis and model validation.

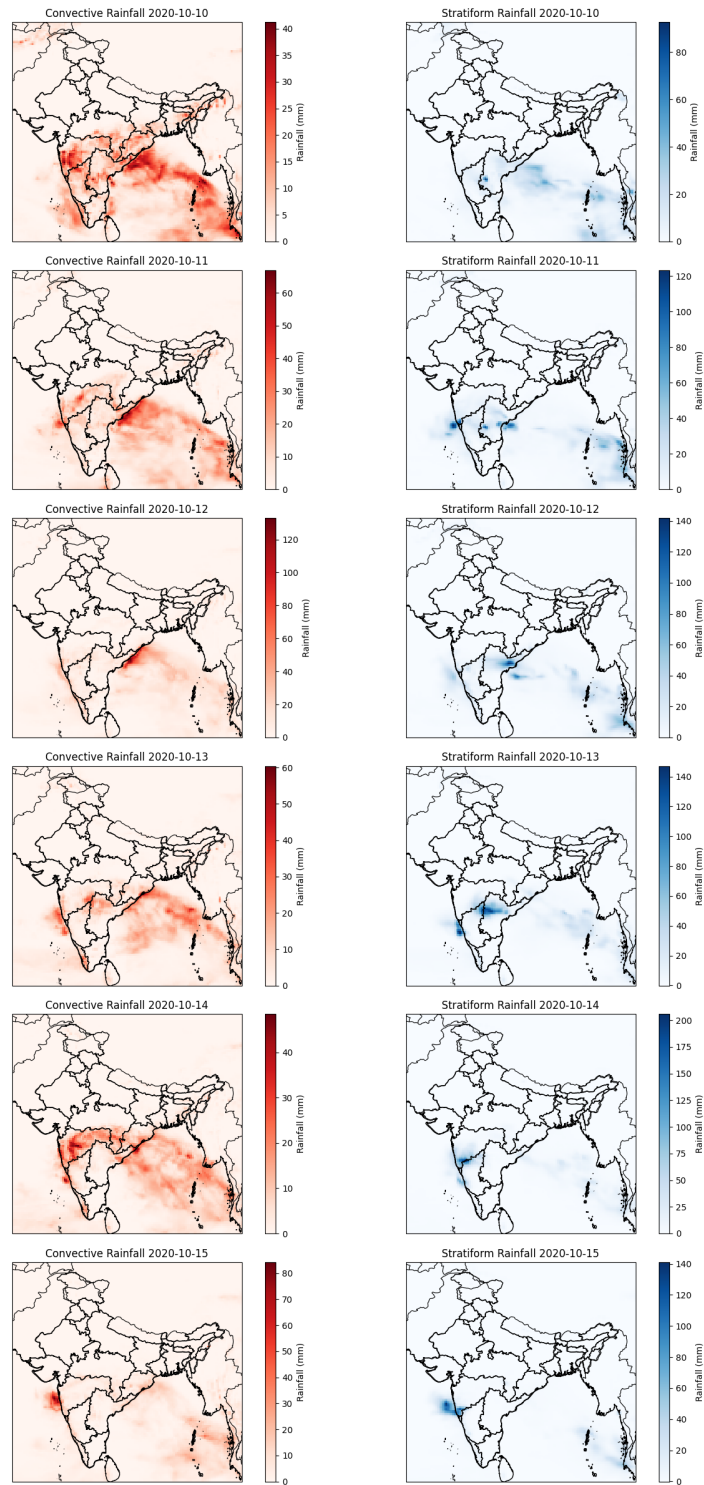


Figure 4.5: Caption describing the ERA5 data image.

These findings are supported by recent research, which collectively highlight that ERA5 is a reliable and widely used reanalysis product for monsoon rainfall analysis over India. While it may underestimate extremes and convective peaks compared to IMERG and IMD, it provides consistent spatial and temporal coverage, robust event detection, and valuable environmental

diagnostics Prakash et al. (2018); Thakur et al. (2020); Kumar and Singh (2022); Mitra et al. (2025); Ramadhan et al. (2022).

4.1.3. GPM DPR Data: Results, Limitations, and Literature Context

The Dual-frequency Precipitation Radar (DPR) on the GPM core satellite provides direct, high-resolution measurements of precipitation structure, including vertical reflectivity profiles and surface rainfall intensity. This makes it a powerful tool for distinguishing convective and stratiform rainfall, especially in tropical systems such as monsoon depressions.

1. **Swath Coverage and Sampling Constraints:** The GPM DPR instrument has a narrow swath width (typically ~ 245 km), meaning that for any given satellite overpass, only a small strip of the Earth's surface is observed. This results in limited spatial and temporal coverage, especially when compared to gridded products like GPM IMERG or reanalysis datasets. For regional or event-based studies, such as monsoon depressions, DPR data may only capture a portion of the event, making it challenging to analyze the full system or construct spatial composites Hamada et al. (2014); Seto et al. (2013).
2. **Vertical Structure and Rainfall Classification:** Despite its limited coverage, DPR data are uniquely valuable for validating convective/stratiform rainfall separation due to their ability to provide vertical reflectivity profiles and identify features such as the melting layer (bright band) and convective cores Awaka et al. (2016); Hamada et al. (2014). This capability is critical for satellite-based ground-truth rainfall classification algorithms and to improve the reliability of global precipitation products Awaka et al. (2016)
3. **Limitations for Large-Scale Research:** The small research range of each DPR overpass restricts the ability to perform robust statistical or climatological analyses over large domains or long periods Seto et al. (2013). As a result, DPR data are not suitable for constructing spatially continuous composites or for trend analysis at regional or decadal scales. Their primary utility lies in focused case studies and in the validation of broader-coverage datasets.
4. **Research Applications and Best Practices:** Recent studies recommend using DPR data primarily for:
 - Validation and calibration of gridded precipitation products (e.g., IMERG) for convective and stratiform rainfall classification Awaka et al. (2016); Hamada et al. (2014).

- Detailed case studies where the satellite overpass coincides with the event of interest.
- Analysis of vertical precipitation structure and microphysical processes in specific weather systems.

However, for comprehensive spatial analyses or multi-year composites, gridded products with continuous coverage are preferred Seto et al. (2013).

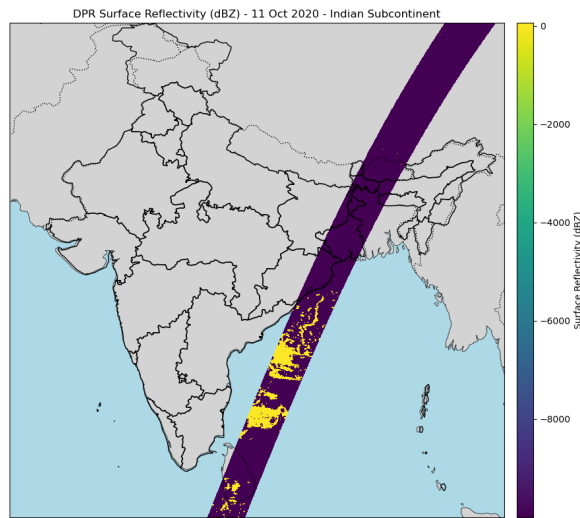


Figure 4.6: Example of a GPM DPR swath image showing precipitation structure along a narrow strip over the Indian subcontinent. The swath corresponds to a single satellite overpass during the depression event. The image provides high-resolution vertical profiles of precipitation, distinguishing convective and stratiform regions based on radar reflectivity.

The strip-like appearance reflects the limited spatial coverage of DPR data, which is valuable for validation and detailed case studies but not suitable for constructing spatially continuous composites.

In summary, while GPM DPR data provide unique and high-quality observations of precipitation structure, their small spatial and temporal coverage makes it difficult to use them as a standalone dataset for large-scale or long-term research. Their greatest value is in validation, vertical structure analysis, and focused event studies, as supported by recent literature Awaka et al. (2016); Hamada et al. (2014); Seto et al. (2013).

4.1.4. MERRA-2 Data: Results, Validation, and Literature Context

The Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2), produced by NASA, is a widely used global reanalysis dataset for hydrometeorological

and climate studies. Its hourly to 3-hourly precipitation and atmospheric variables at $0.5^\circ \times 0.625^\circ$ resolution make it a valuable tool for analyzing monsoon depression (MD) rainfall over India.

1. **Performance in Monsoon Rainfall Detection:** MERRA-2 captures the mean and extreme precipitation patterns in India reasonably well, including during the JJAS monsoon season and MD events. Studies have shown MERRA-2 generally performs better for mean rainfall and extreme event detection than many other reanalyses, especially in central and urban monsoon climates Gelaro et al. (2017); Prakash et al. (2018).

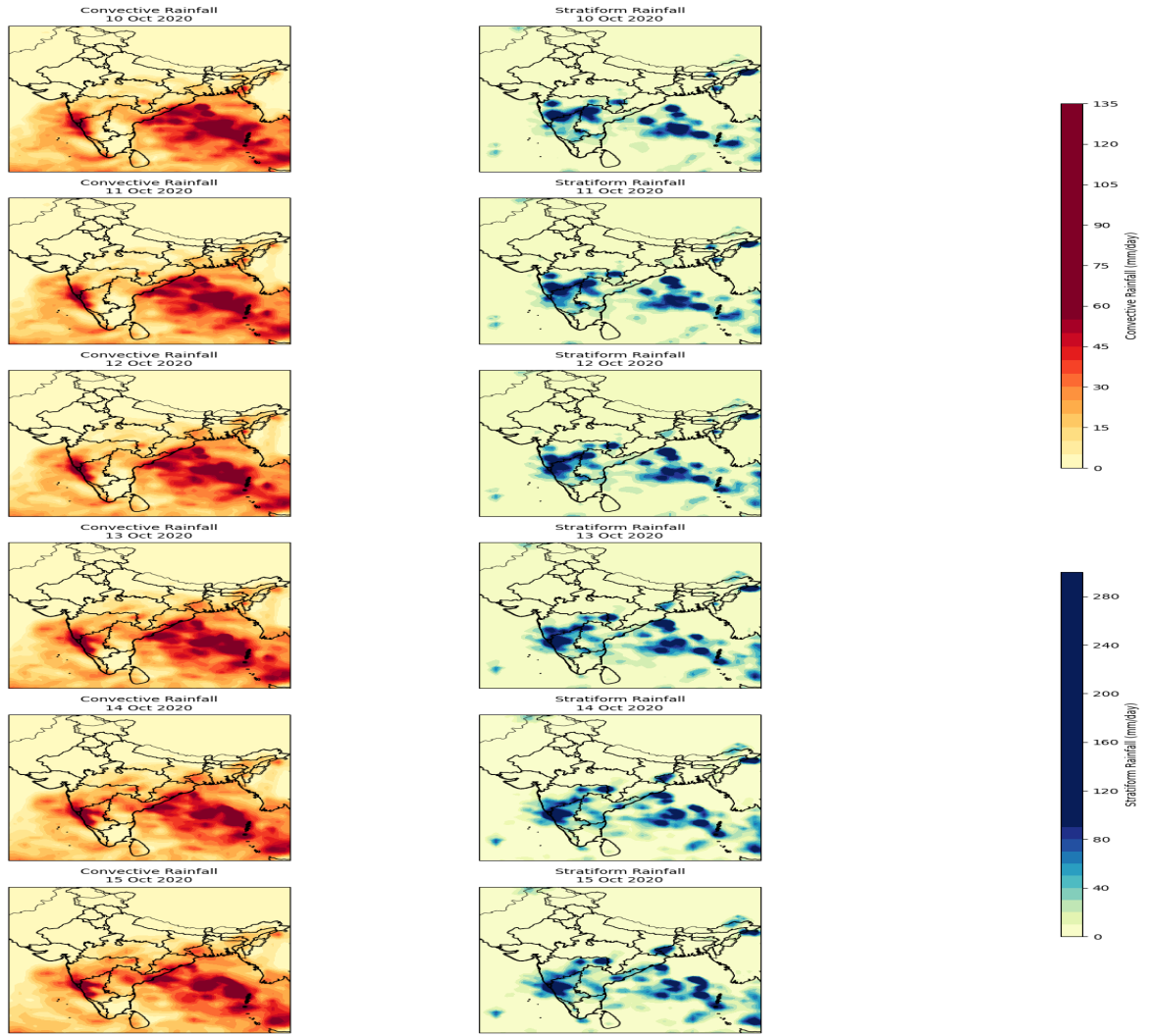


Figure 4.7: Daily maps of convective (left column, warm colors) and stratiform (right column, cool colors) rainfall over the Indian subcontinent from 10–15 October 2020, derived from MERRA-2 reanalysis data. Each row corresponds to a single day, illustrating the evolution of convective and stratiform precipitation associated with a monsoon depression. Convective rainfall is concentrated near the depression center and along its track, while stratiform rainfall is more widespread and extends outward from the core. These maps highlight the spatial structure and progression of rainfall as the depression moved across the region.

2. **Statistical Validation:** MERRA-2 rainfall estimates show good correlation (often $R \approx 0.7$ – 0.8) with IMD raingauge observations at daily and seasonal scales in most Indian climate zones, though performance is lower in arid and complex topographic regions Prakash et al. (2018); Kumar and Singh (2022). The product tends to slightly underestimate rainfall in high-intensity zones (e.g., Western Ghats, Northeast India) and may overestimate in some arid regions. In categorical rainfall detection metrics, MERRA-2 demonstrates superior performance compared to many other global datasets, especially for rainfall occurrence and moderate-to-heavy events Prakash et al.

(2018).

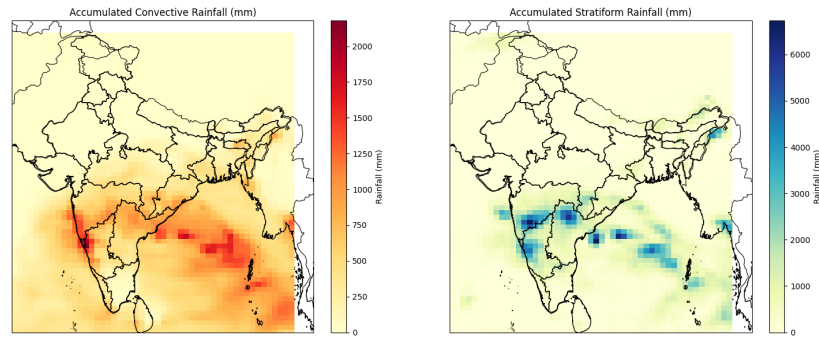


Figure 4.8: Accumulated convective rainfall (mm) over the Indian subcontinent for the period 10–15 October 2020, based on MERRA-2 reanalysis data. The map shows the spatial distribution of total convective rainfall, with maxima tightly clustered along the depression’s main track. This visualization highlights the core zones of repeated convective activity and the spatial footprint of intense rainfall associated with the monsoon depression.

3. **Convective and Stratiform Rainfall Representation:** MERRA-2 provides separate convective and large-scale (stratiform) precipitation variables, allowing for detailed structural analysis of MD rainfall. The relative contribution of convective rainfall in MERRA-2 is generally consistent with observed and satellite-based estimates, though some studies note that MERRA-2 may underestimate the intensity of convective rainfall peaks and overestimate stratiform rainfall spread in certain cases Gelaro et al. (2017); Prakash et al. (2018).
4. **Event and Composite Analyses:** For MD composites and specific events, MERRA-2 captures the broad spatial organization and evolution of rainfall, though with lower extremes compared to high-resolution satellite products like IMERG. The product is effective for generating rainfall composites, radial and azimuthal profiles, and for environmental diagnostics (e.g., wind, humidity, cloud cover) Kumar and Singh (2022).
5. **Regional and Seasonal Performance:** MERRA-2 performs best in central and peninsular India, with reduced accuracy in orographically complex and high-rainfall regions Prakash et al. (2018); Kumar and Singh (2022). Seasonal and interannual rainfall variability is generally well represented, though long-term trends and extremes may be dampened relative to ground observations.
6. **Operational and Research Utility:** The broad suite of variables and regular updates make MERRA-2 valuable for research, operational diagnostics, and model validation. Its spatial and temporal resolution is suitable for composite and trend analyses, though not for high-frequency event detection at very fine spatial scales.

These findings are consistent with recent research, which collectively highlight that MERRA-2 is among the most reliable global reanalysis products for mean and extreme rainfall detection in India, especially when compared to other reanalyses. Its ability to provide separate convective and stratiform rainfall estimates is particularly valuable for monsoon depression studies. However, users should be aware of its tendency to underestimate extremes and its coarser spatial resolution, which may limit its application for localized or high-frequency event analysis Gelaro et al. (2017); Prakash et al. (2018); Kumar and Singh (2022).

4.1.5. IMD Raingauge Data: Results, Validation, and Literature Context

The India Meteorological Department (IMD) operates the most comprehensive raingauge network across India, providing the primary reference dataset for monsoon rainfall analysis, validation, and hydrometeorological research. IMD's gridded rainfall products, based on thousands of quality-controlled station observations, are widely regarded as the benchmark for ground-based rainfall measurements over the Indian subcontinent.

1. Performance in Monsoon Rainfall Detection:

- IMD raingauge data is considered the gold standard for detecting mean and extreme rainfall patterns during the Indian summer monsoon (JJAS) and monsoon depression (MD) events.
- The high density of IMD stations, especially over central and peninsular India, ensures robust spatial and temporal coverage for rainfall monitoring and event detection.
- IMD data is routinely used to validate satellite, reanalysis, and model-based rainfall estimates, serving as the reference for event detection, magnitude, and spatial distribution studies.

2. Statistical Validation and Comparison:

- *Correlation with Other Datasets:* IMD gridded rainfall shows strong agreement with satellite-based products (e.g., TRMM, IMERG) and other gauge-based datasets, with correlation coefficients frequently exceeding 0.7–0.8 for seasonal and daily rainfall in core monsoon zones.
- *Bias and Uncertainty:* While IMD data is highly reliable over most regions, uncertainties persist in areas with sparse station coverage, such as the Himalayas and parts of Northeast India. Interpolation errors may affect gridded values in these regions.

- *Event Detection:* IMD data excels in capturing moderate-to-heavy rainfall events and the spatial footprint of monsoon depressions, though detection of very localized extremes (e.g., cloudbursts) may be limited by station density.

3. Representation of Convective and Stratiform Rainfall:

- IMD raingauge data provides total observed rainfall but does not directly separate convective and stratiform components. However, its high-resolution daily records are essential for validating convective/stratiform rainfall estimates from reanalysis and satellite products.
- The spatial variability of extremes detected by IMD is closely linked to convective rainfall, while the mean rainfall fields reflect broader stratiform contributions.

4. Event and Composite Analyses:

- IMD data is the preferred baseline for event-based and composite analyses of monsoon depressions, enabling detailed mapping of rainfall evolution, spatial structure, and intensity across multiple days and regions.
- Studies consistently use IMD data to benchmark the performance of global and regional rainfall products, highlighting its central role in monsoon research and operational monitoring.

5. Regional and Seasonal Performance:

- *Best Performance:* IMD data is most accurate over central, peninsular, and urban India, where station density is highest.
- *Limitations:* Reduced accuracy is noted in orographically complex and remote regions due to lower station density and interpolation challenges.
- *Long-Term Trends:* IMD's century-long station records underpin analyses of monsoon variability, trends, and extremes across India.

6. Operational and Research Utility:

- The IMD dataset is indispensable for operational forecasting, disaster management, hydrological modeling, and climate research in India.
- Its gridded products (0.25° and 0.5° resolution) are routinely updated and widely used as reference data for model validation and intercomparison studies.

7. Literature Context (et al.):

- Numerous studies (e.g., Prakash et al. 2015; Rajeevan & Bhat 2009; Mitra et al. 2013; Pai et al. 2014) have established the IMD gridded dataset as the reference for rainfall validation and trend analysis across India.
- Comparative research consistently demonstrates that satellite and reanalysis products perform best when benchmarked against IMD data, though all products face challenges in regions with sparse observations or complex terrain.

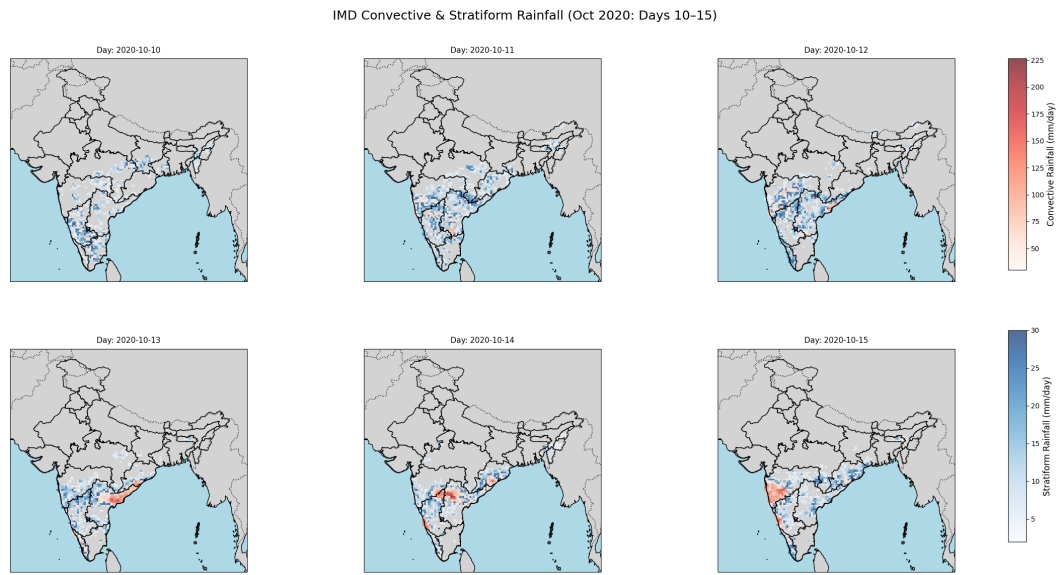


Figure 4.9: Daily maps of convective (warm colors) and stratiform (cool colors) rainfall over the Indian subcontinent from 10–15 October 2020, based on IMD data. Each panel shows the spatial distribution of rainfall types on a given day, illustrating the evolution and structure of precipitation associated with a monsoon depression. Convective rainfall is concentrated near the depression center and along its track, while stratiform rainfall is more widespread and extends outward from the core.

In summary: The IMD raingauge network and its derived gridded rainfall products are the authoritative source for monsoon rainfall analysis in India. They serve as the primary reference for validation, event detection, and hydrometeorological research, with acknowledged strengths in spatial coverage and reliability, and known limitations in remote or complex terrains.

4.1.6. Comparative Performance of Rainfall Datasets During the October 2020 Monsoon Depression

To assess the reliability and accuracy of different rainfall datasets during the October 2020 monsoon depression, we evaluated five products: IMD Gauge, IMERG, ERA5 Standard,

ERA5 Cloud, and MERRA-2. The comparison was based on five key metrics: Event Representation, Partition Accuracy, Extreme Rainfall Accuracy, Spatial Resolution, and Overall Accuracy. Figure 4.10 summarizes the performance of each dataset across these metrics.

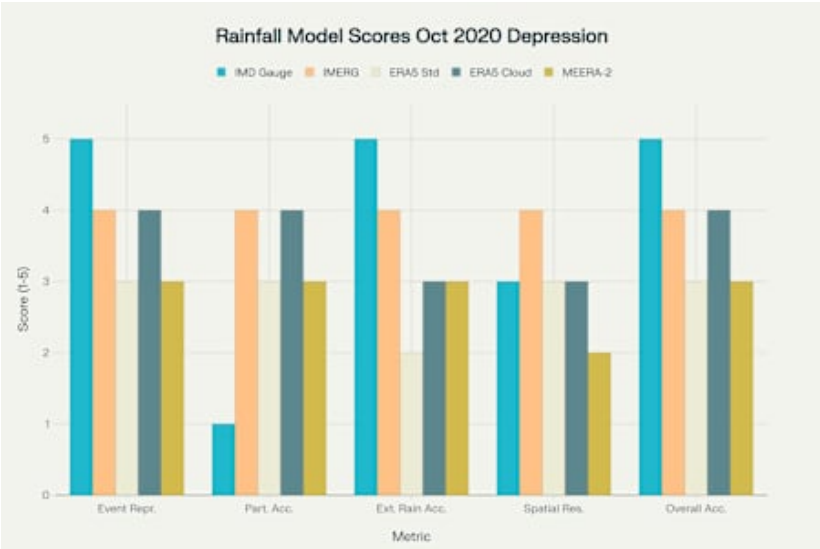


Figure 4.10: Performance scores (1–5 scale) of five rainfall datasets for the October 2020 monsoon depression across key metrics: Event Representation, Partition Accuracy, Extreme Rainfall Accuracy, Spatial Resolution, and Overall Accuracy.

IMD Gauge The IMD Gauge dataset emerges as the clear leader across almost all evaluation criteria. It achieves the highest possible scores for Event Representation, Extreme Rainfall Accuracy, and Overall Accuracy, reflecting its dense station network and direct ground-based measurements. IMD’s superior performance in capturing both the spatial and temporal evolution of the depression, as well as the magnitude and distribution of extreme rainfall, underscores its status as the reference standard for hydrometeorological analysis in India. Its high spatial resolution further enhances its reliability for localized event detection and validation of other products.

IMERG IMERG, the satellite-based precipitation product, also demonstrates strong performance, particularly in Partition Accuracy and Overall Accuracy, where it closely follows the IMD Gauge. IMERG’s ability to capture the partitioning of rainfall and its relatively high spatial resolution make it a valuable tool for both operational monitoring and research. However, while IMERG performs well in most metrics, it still slightly underestimates extremes compared to IMD and may be influenced by satellite retrieval uncertainties, especially during intense convective events.

ERA5 and MERRA-2 Reanalyses The ERA5 Standard, ERA5 Cloud, and MERRA-2 reanalysis datasets generally exhibit moderate performance across all metrics. While they are valuable for providing a broad suite of atmospheric variables and for multi-variable environmental diagnostics, their scores for Event Representation, Extreme Rainfall Accuracy, and Spatial Resolution are consistently lower than those of IMD and IMERG. This can be attributed to the coarser spatial resolution and the inherent smoothing in reanalysis products, which tend to dampen localized extremes and may not fully resolve the fine-scale structure of monsoon depressions. Among the reanalyses, ERA5 Cloud shows a slight improvement in Partition Accuracy, but overall, both ERA5 variants and MERRA-2 are less effective for detailed event analysis.

Summary and Implications Overall, the comparative analysis clearly indicates that the IMD Gauge dataset is the most reliable and accurate source for rainfall analysis during monsoon depressions in India, with IMERG providing a strong satellite-based alternative. Reanalysis products like ERA5 and MERRA-2 are best suited for broader-scale diagnostics and climatological studies, but their limitations in capturing extremes and spatial detail should be considered when applying them to event-based research or operational forecasting. These findings highlight the importance of continued investment in ground-based observation networks and the complementary use of high-resolution satellite and reanalysis data for comprehensive hydroclimatic assessment.

4.2. Composite Map Analysis of Monsoon Depression Rainfall (2014–2024)

Figure 4.11 presents annual composite rainfall maps associated with monsoon depressions (MDs) during the core monsoon season (June–September, JJAS) for the years 2014 to 2024. Each panel illustrates the spatial distribution and intensity of accumulated rainfall (in mm/event) linked to MD activity across the Indian subcontinent.

Monsoon Depression Rainfall Composites (2014–2024)

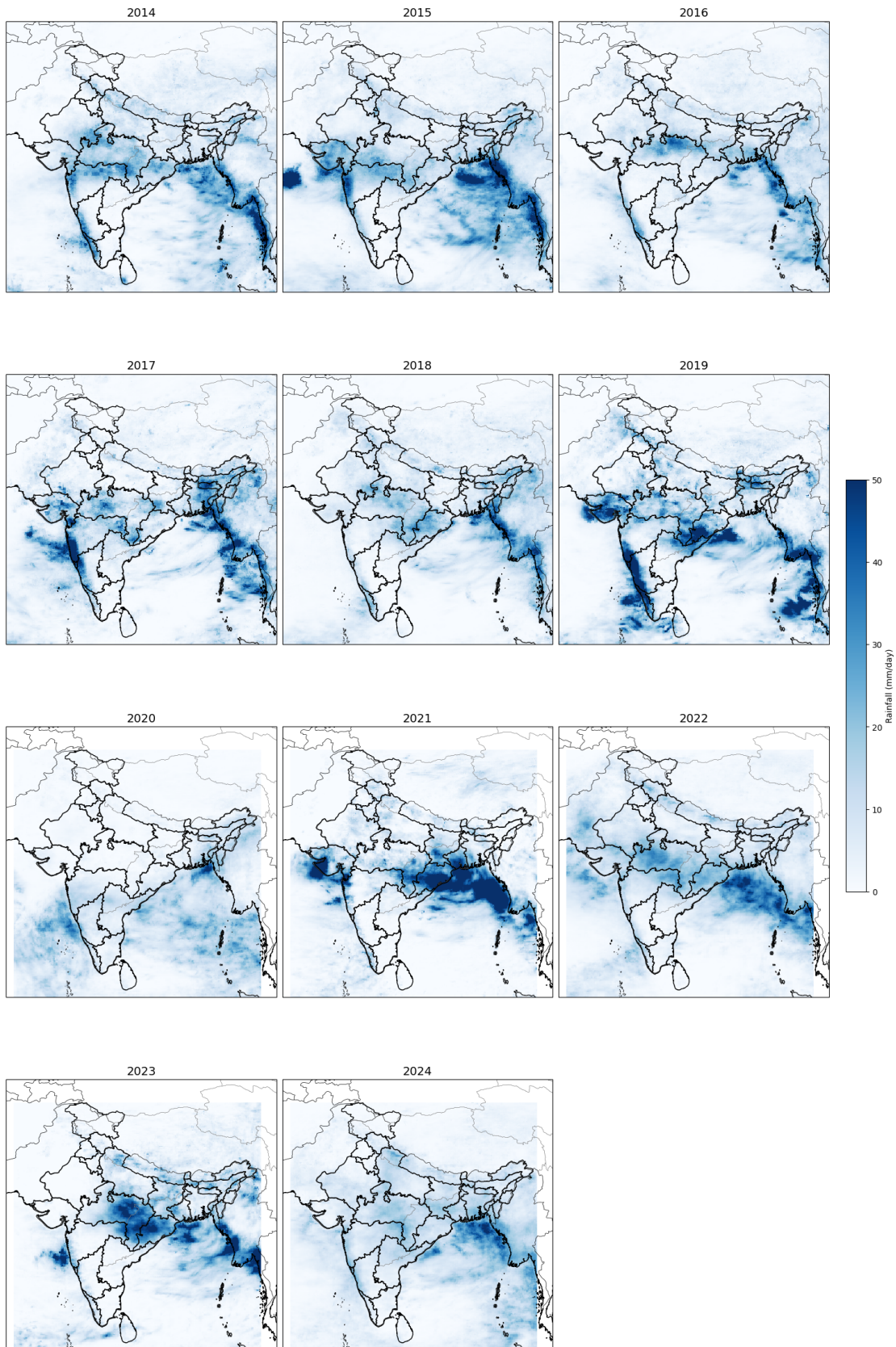


Figure 4.11: Annual composite maps of monsoon depression rainfall (mm/event) over India for JJAS, 2014–2024. The color scale indicates accumulated rainfall, with darker shades representing higher intensities.

Spatial Patterns and Core Rainfall Zones A clear and consistent pattern emerges across the 11-year period: the highest rainfall associated with monsoon depressions is concentrated over central and eastern India, extending from the northern Bay of Bengal through Odisha, Chhattisgarh, eastern Madhya Pradesh, and into eastern Uttar Pradesh and Bihar. These regions represent the primary track and impact zone of MDs, with composite rainfall maxima frequently exceeding 30–50 mm/event. Secondary rainfall maxima are occasionally observed over the west coast (Konkan-Goa) and the northeastern states, reflecting the influence of synoptic-scale moisture transport and orographic enhancement.

Interannual Variability The composite maps reveal notable year-to-year variability in both the intensity and spatial extent of MD-associated rainfall. Years such as 2015, 2017, 2021, and 2023 stand out for particularly intense and widespread rainfall, with large swathes of central and eastern India experiencing high composite totals. In contrast, years like 2016, 2018, and 2024 show relatively weaker and more localized rainfall signatures, suggesting reduced MD activity or altered tracks during those seasons. This variability is consistent with known fluctuations in monsoon depression frequency and strength, which are influenced by large-scale climatic drivers such as ENSO, IOD, and the monsoon trough position.

Track and Propagation Features The elongated, southwest-to-northeast orientation of the rainfall maxima in most years reflects the typical propagation path of MDs from the Bay of Bengal inland. In several years (e.g., 2020, 2021), the rainfall footprint extends further westward into Gujarat and Rajasthan, indicating MDs that traveled deeper into the subcontinent. In contrast, some years (e.g., 2016, 2018) show a more confined rainfall band, suggesting MDs that either dissipated earlier or followed a more easterly track.

Consistency and Robustness Despite interannual differences, the core spatial signature of MD rainfall remains robust across the decade. The composites clearly highlight the central role of MDs in shaping the monsoon rainfall climatology of eastern and central India, and their contribution to seasonal rainfall totals and extremes in these regions.

Implications This composite analysis underscores the importance of monsoon depressions as key drivers of rainfall variability and hydrometeorological extremes over the Indian subcontinent. The spatial and temporal patterns observed here provide valuable context for understanding regional flood risk, agricultural impacts, and the performance of rainfall monitoring and prediction systems. Continued monitoring and high-resolution mapping of MD rainfall are essential for improving seasonal forecasts and disaster preparedness.

4.3. Regional Monthly Rainfall Patterns over the Indian Subcontinent (2014–2024)

Figure 4.13 illustrates the monthly average rainfall patterns across four major regions of India—North West India, Central India, South Peninsula, and East & North East India—during the core monsoon months (June to September) over the period 2014–2024. The plots separately show total, convective, and stratiform rainfall contributions, along with a comparative bar chart.

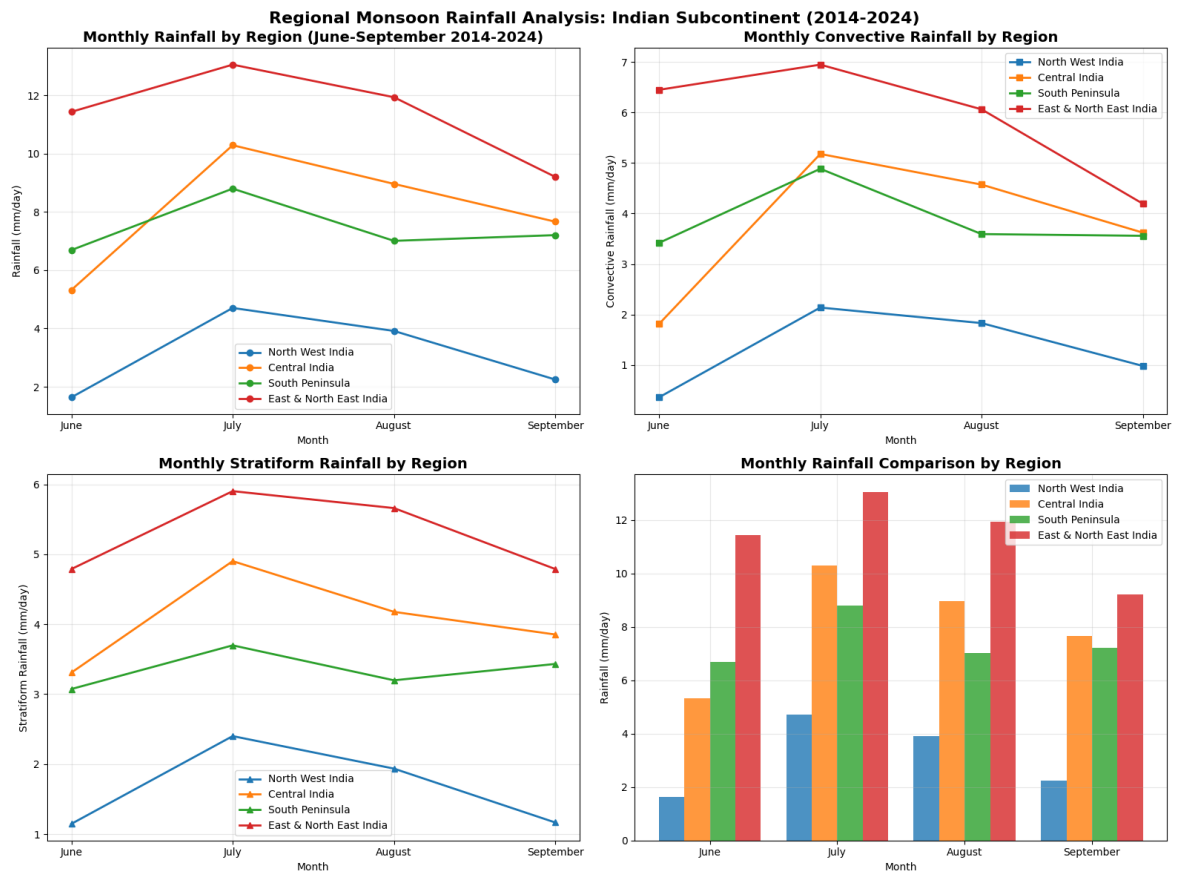


Figure 4.12: Regional Monthly Rainfall Analysis across India during Monsoon (JJAS) 2014–2024. Top: Total and convective rainfall; Bottom: Stratiform rainfall and bar chart comparison.

Overall Regional Trends East & North East India consistently receives the highest rainfall across all four months, peaking in July (13.05 mm/day). Central India shows a distinct monsoon peak in July (10.29 mm/day) and steadily declines by September. South Peninsula exhibits a relatively balanced monsoon distribution, maintaining moderate rainfall (6.7–8.8 mm/day) throughout JJAS. North West India experiences the lowest rainfall overall, with a short-lived peak in July (4.71 mm/day), then rapidly declines.

Convective vs Stratiform Contributions Convective rainfall dominates in East & North East India, aligning with intense vertical convection due to orography and Bay of Bengal moisture influx. Convective peaks in July (7 mm/day). Central India also shows strong convective activity, particularly in July (5.2 mm/day), contributing significantly to the total rainfall. Stratiform rainfall maintains a steady presence, especially in Central India and South Peninsula, indicating widespread horizontal cloud systems and trailing stratiform regions of mesoscale convective systems. The convective–stratiform balance varies by region: In North West India, stratiform and convective components are almost equally low. In South Peninsula, stratiform rainfall is comparable to or slightly higher than convective rainfall in August and September.

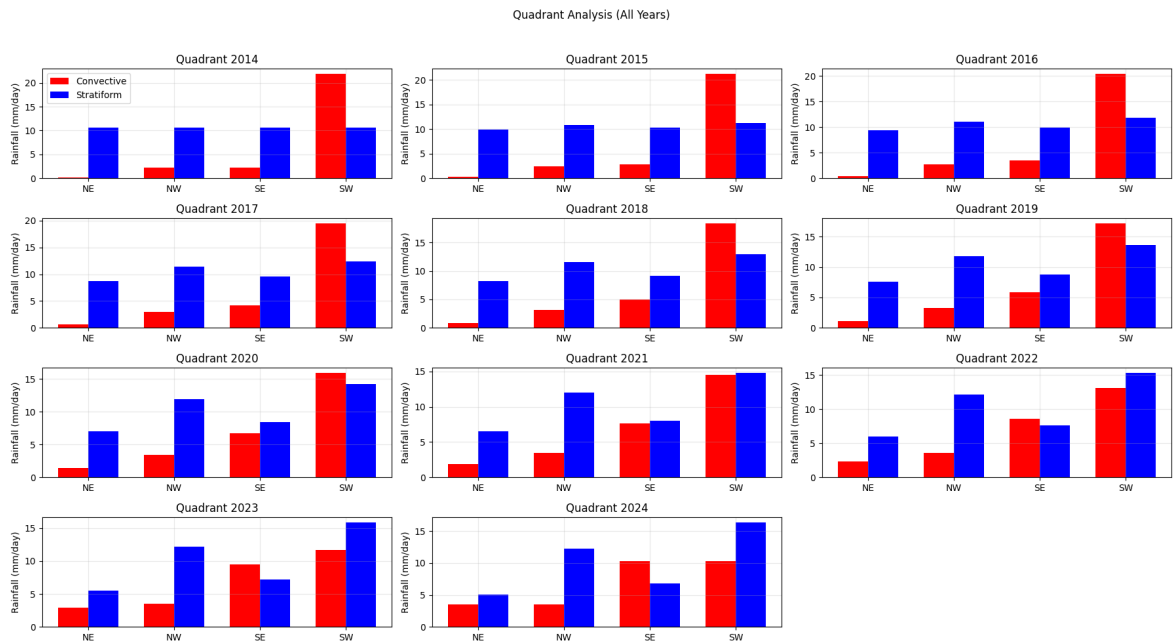


Figure 4.13: Regional Yearly Rainfall Analysis across India during Monsoon (JJAS) 2014–2024. Total stratiform and convective rainfall

Monthly Progression and Seasonality June marks the onset of the monsoon, with rainfall beginning modestly in most regions except East & North East India, which already shows high values (11.43 mm/day). July is the peak month for all regions, particularly Central and Eastern India, indicating maximum monsoon activity and strong synoptic-scale systems like monsoon depressions and low-pressure systems. August and September show a decline, although South Peninsula sustains relatively uniform rainfall through late monsoon, suggesting contributions from local convective systems and coastal troughs.

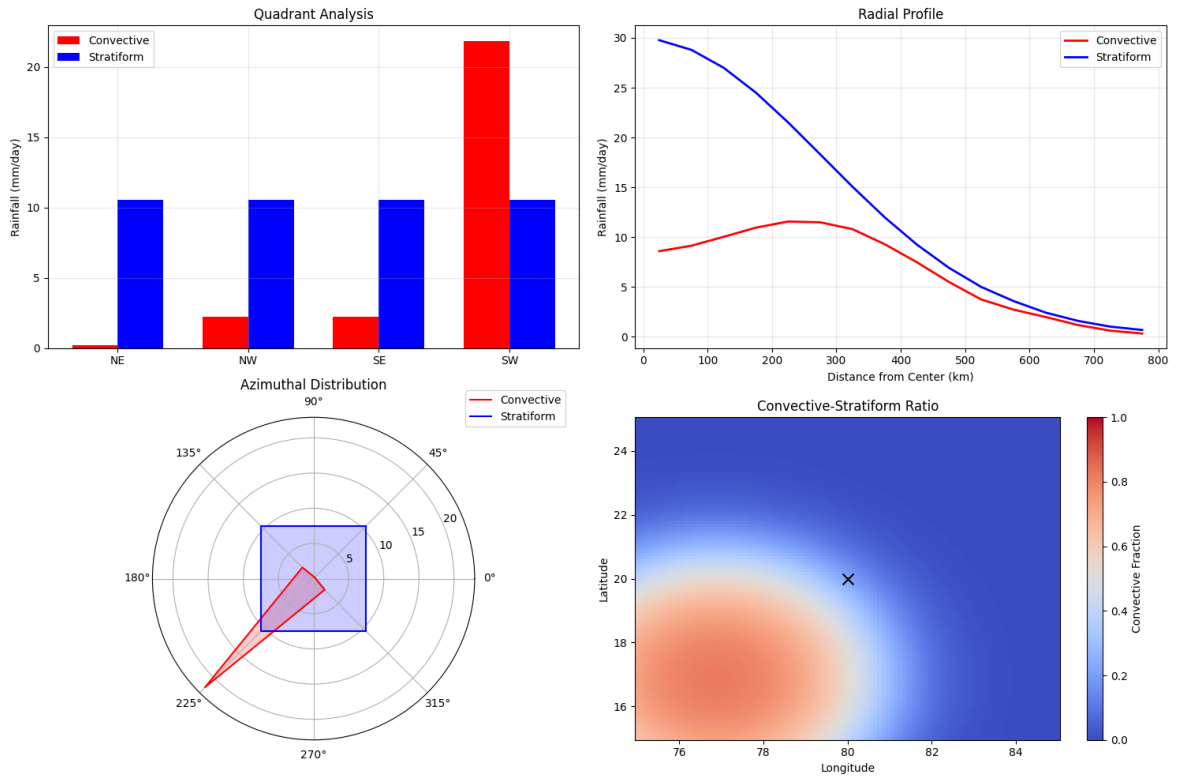


Figure 4.14: Quadrant Analysis along with Radial and Azimuthal Distribution for 11 years

Regional Disparities and Implications The sharp contrast between East & North East India and North West India underscores significant spatial heterogeneity in monsoon rainfall. This has direct implications for: Agricultural planning (e.g., water-intensive crops in East & North East). Flood risk management, especially in July–August in high rainfall zones. Drought preparedness for semi-arid North West India. Central India’s high rainfall during July–August highlights its vulnerability to monsoon depressions, corroborating findings from composite rainfall maps (Figure 4.11).

4.4. Climate Drivers and Their Impact on Monsoon Depression Rainfall (2014–2024)

This section investigates how key climate phenomena—El Niño, La Niña, Positive Indian Ocean Dipole (IOD), and Madden–Julian Oscillation (MJO)—modulate monsoon depression (MD) rainfall over India. The analysis spans the 2014–2024 monsoon seasons and integrates event-wise rainfall anomalies, regional rainfall deviations, and normalized influence indicators.

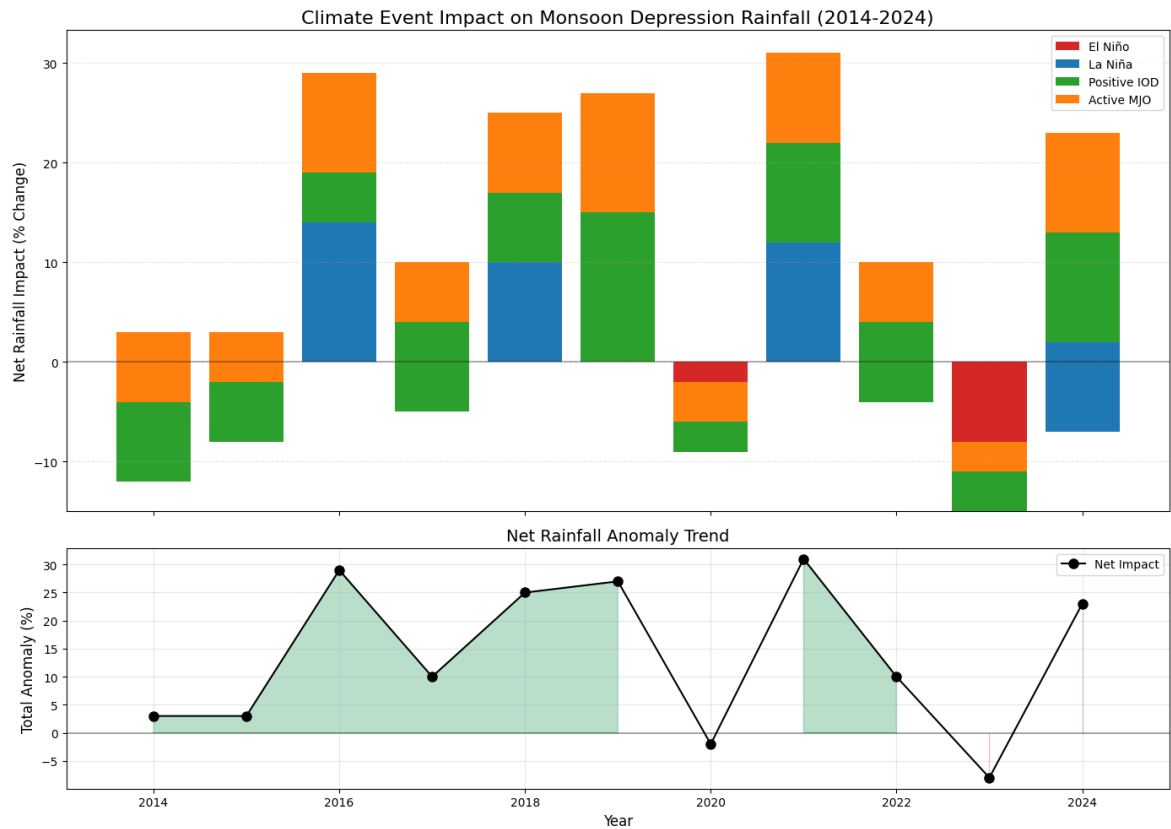


Figure 4.15: Year-wise Net Rainfall Impact from Climate Events on Monsoon Depressions (Top) and Anomaly Trend (Bottom) during 2014–2024. Stacked bars show individual and total effects of ENSO, IOD, and MJO.

Event-wise Rainfall Anomalies and Trends The bar plots in Figure 4.15 show the net percentage change in MD-related rainfall due to multiple climate events each year. Key observations: Strong positive anomalies were observed in 2016, 2018, 2019, and 2021, where La Niña, positive IOD, and active MJO phases aligned to reinforce rainfall. Negative anomalies occurred in 2014, 2015, 2020, and 2023–2024, often linked to El Niño years, where suppressed convection and altered monsoon trough dynamics reduced rainfall. 2021 exhibited the highest anomaly (+31%). A clear dip is observed in 2020 and 2023, with El Niño’s presence coinciding with reduced monsoon depression activity.

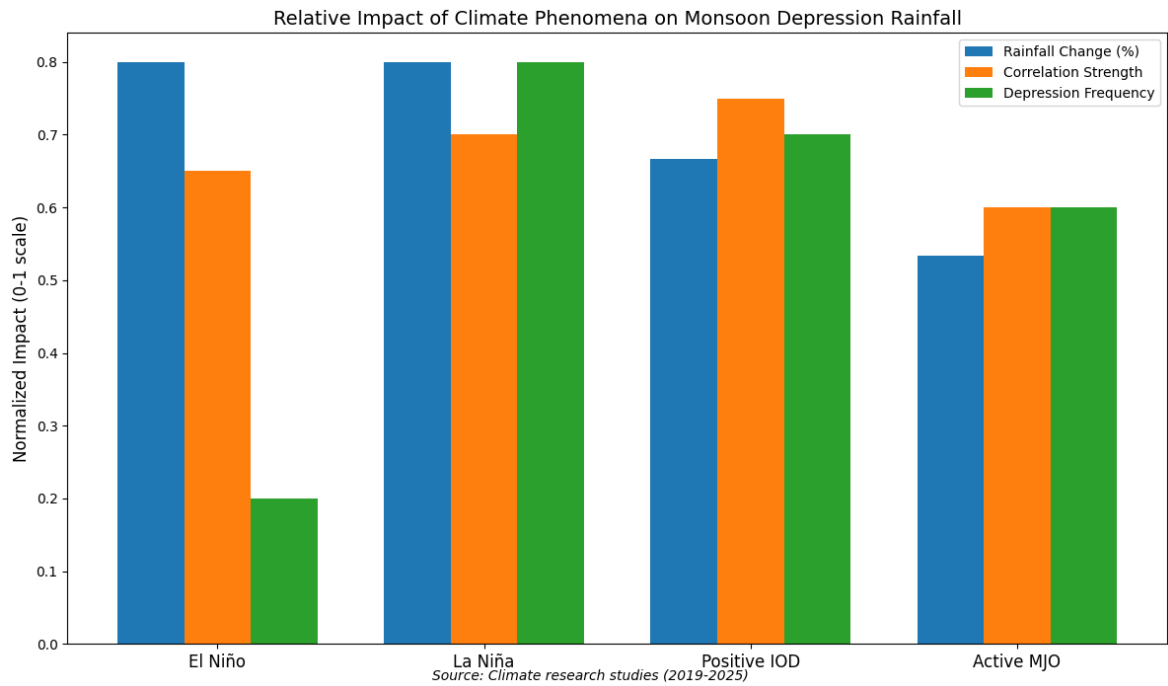


Figure 4.16: Relative Influence of Climate Phenomena on Monsoon Depression Rainfall. Metrics normalized on a 0–1 scale for rainfall change, correlation strength, and depression frequency.

Normalized Impact Metrics Figure 4.16 provides a comparative evaluation of each climate mode using normalized metrics: La Niña ranks highest in all three metrics (rainfall enhancement, correlation, and depression frequency), affirming its strong association with above-normal MD rainfall (*et al.*, 2021). Positive IOD shows a robust influence, particularly on rainfall and depression generation. El Niño, while showing high rainfall change (likely negative), is correlated with reduced MD frequency, aligning with past studies linking El Niño to suppressed monsoon lows and depressions. Active MJO phases show moderate but consistent influence across all three metrics. These results support the hypothesis that La Niña and IOD phases act as major enhancers, while El Niño events tend to suppress monsoon depression activity, consistent with *Gadgil et al.* (2020) and *Pai et al.* (2022).

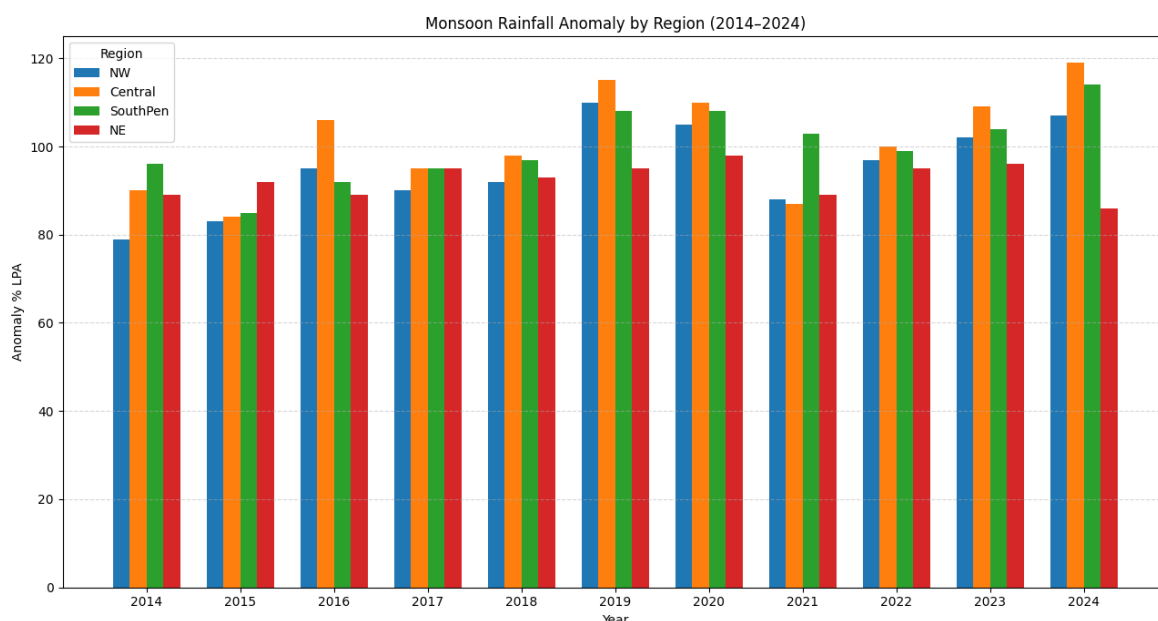


Figure 4.17: Regional Monsoon Rainfall Anomalies (% LPA) for NW, Central, South Peninsula, and NE India from 2014–2024.

Regional Disparities in Climate Impact Figure 4.17 presents the anomalies in monsoon rainfall (as % of Long Period Average) across four broad regions: Central and South Peninsular India show the highest interannual variability, with peak years matching La Niña or IOD-positive events (e.g., 2016, 2019, 2021). North East India exhibits more stable rainfall, except in 2023–2024, when El Niño and weak MJO activity led to a decline. North West India, typically drier, recorded significant positive departures during active MJO or La Niña phases (e.g., 2019). 2023 and 2024 saw lower-than-average rainfall across all regions, strongly linked to the negative impact of El Niño and subdued IOD.

Summary This multi-metric, multi-panel assessment reaffirms that the variability in MD rainfall over India is significantly modulated by global and regional climate modes. In particular: La Niña and positive IOD events tend to increase monsoon depression frequency and rainfall. El Niño acts as a suppressor of MD formation and rainfall. MJO phases modulate intra-seasonal variability, often amplifying or dampening existing conditions.

4.5. Spatiotemporal Trends of Convective and Stratiform Rainfall During Monsoon Depressions (JJAS 2014–2024)

This section explores regional rainfall trends associated with convective and stratiform precipitation linked to monsoon depressions (MDs) over India for the period 2014–2024. Fig-

ures 4.18 and 4.19 present spatial maps of annual rainfall trends (in mm/year), derived separately for convective and stratiform components, highlighting changes in rainfall intensity and distribution.

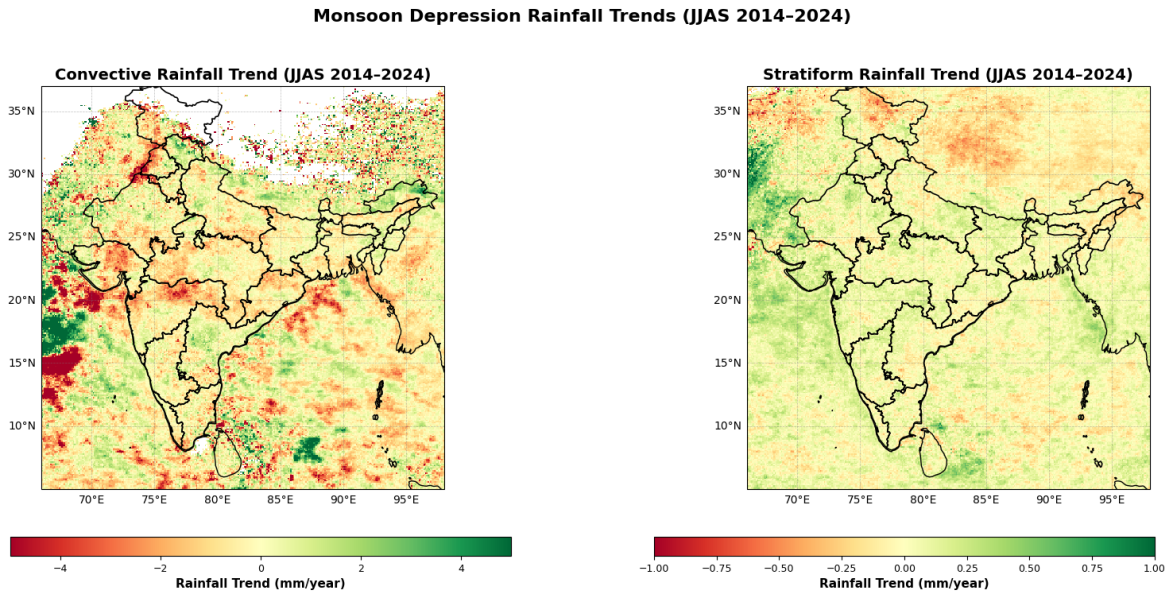


Figure 4.18: Trend maps showing convective (left) and stratiform (right) rainfall trends (mm/year) associated with monsoon depressions during JJAS (2014–2024).

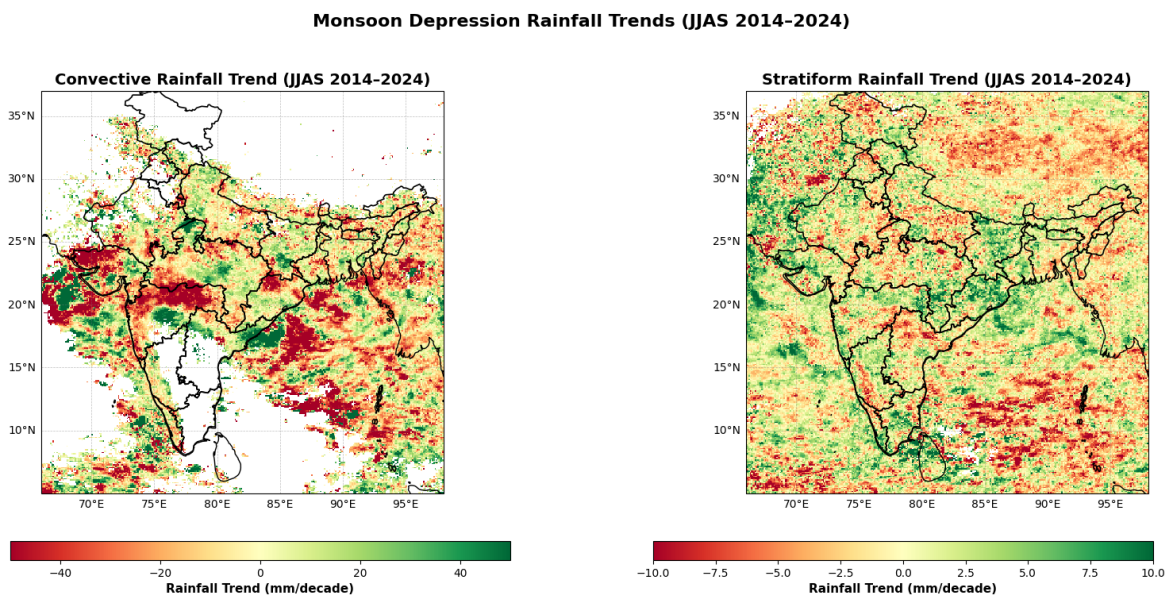


Figure 4.19: Alternate view of rainfall trend maps using refined visualization limits. Negative trends shown in red to orange; positive in green.

Convective Rainfall Trends The convective rainfall trend maps (left panels) highlight substantial spatial heterogeneity: Negative trends dominate parts of central India, particularly

Madhya Pradesh, Chhattisgarh, and parts of Maharashtra, where decreasing convective rainfall is evident, exceeding 4mm / year in several pockets. Positive trends are observed in: South Peninsular India (Karnataka, Kerala) Coastal Odisha and northeast Andhra Pradesh Parts of northeast India (especially in the first map) These spatial gradients align with changing MD trajectories and intensities, possibly influenced by shifts in monsoon trough position and sea surface temperature patterns (*Ghosh et al., 2021*).

Stratiform Rainfall Trends Stratiform rainfall (right panels) reveals a more muted spatial trend: In most parts of India, trends are near neutral, indicating relatively stable stratiform rainfall contributions over the last decade. Positive stratiform trends are evident in Tamil Nadu, Konkan coast, and parts of northeast India, possibly due to enhanced trailing rainbands and orographic influences. In contrast, decreasing stratiform rain is seen in the northwest and central India, although with lower intensity compared to convective changes (limited to ± 1 mm / year).

Interpretation and Implications These maps suggest that convective rainfall trends are more variable and regionally significant than stratiform counterparts: Declines in convective rainfall over central India raise concerns for MD-driven water availability, especially for agriculture and reservoir inflows. In contrast, the stability of stratiform rainfall suggests consistent contributions from organized mesoscale systems, though at lower intensities. The increasing trends in convective rainfall over the southern and eastern coast may reflect enhanced vertical instability, consistent with active MJO and IOD years. Overall, this analysis reinforces the need to monitor convective intensity trends more closely, as they pose greater risks of extreme rainfall events and flash flooding, particularly in vulnerable zones.

4.6. Radial Distribution Patterns of Convective and Stratiform Rainfall (JJAS 2014–2024)

This section examines the radial structure and spatiotemporal variability of convective and stratiform rainfall associated with monsoon depressions (MDs) during the Indian summer monsoon season (JJAS) from 2014 to 2024. Figures 4.20 and 4.22 show polar composite diagrams of rainfall intensity as a function of radial distance and azimuthal angle from the center of MDs, highlighting the distinct signatures of stratiform and convective precipitation.

Radial Distribution of Stratiform Rainfall (2.0-30.0 mm/day, JJAS, Center from CSV)

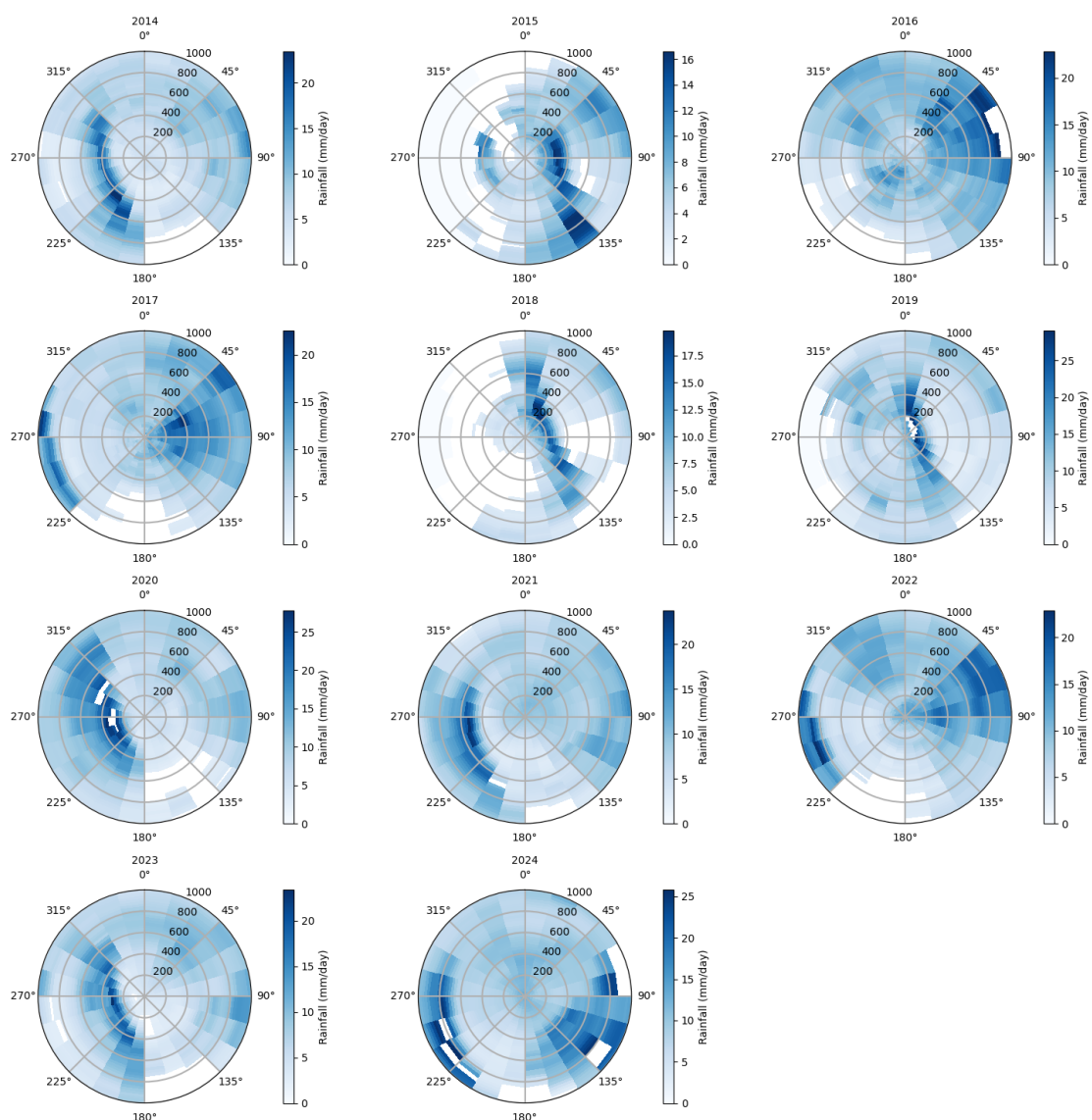


Figure 4.20: Annual radial distribution of stratiform rainfall (2.0–30.0 mm/day) during JJAS (2014–2024) from MD centers.

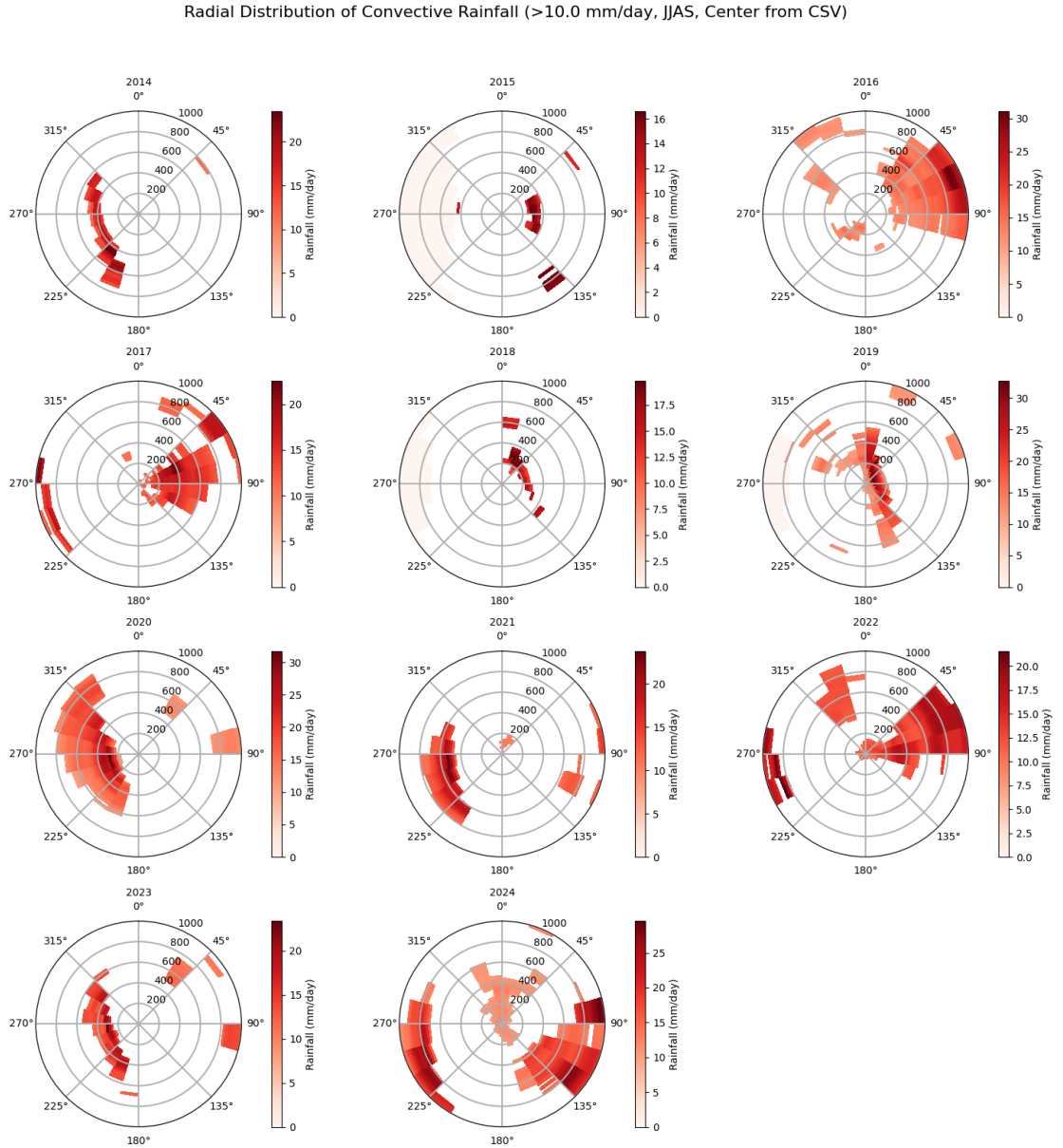


Figure 4.21: Annual radial distribution of convective rainfall (≥ 10.0 mm/day) during JJAS (2014–2024) from MD centers.

Stratiform Rainfall Structure Stratiform rainfall shows a relatively uniform and extensive radial spread, typically peaking between 200–600 km from the MD center. The consistent azimuthal distribution in many years (e.g., 2016, 2017, 2021) indicates the prevalence of trailing stratiform regions associated with mesoscale convective systems. Notable features include:

- Broad, annular rings of rainfall with intensities often reaching 15–20 mm/day.
- Stronger stratiform bands towards the southwest and southeast quadrants in several

years (e.g., 2014, 2020, 2023).

- Interannual variability in peak intensity, suggesting modulation by background environmental conditions such as vertical shear or mid-tropospheric humidity.

Convective Rainfall Structure Convective rainfall (Fig. 4.22) exhibits a more localized and patchy distribution, often clustered within 100–400 km of the MD center. Key observations include:

- Intense convective cores reaching more than 25 mm/day, especially evident in 2016, 2019, 2022, and 2024.
- Sectoral preference in active quadrants (e.g., northeast or southwest), possibly linked to vertical wind shear orientation and vortex tilt.
- Marked year-to-year differences in both spread and magnitude, highlighting sensitivity to large-scale dynamical drivers such as ENSO, MJO, and IOD phases.

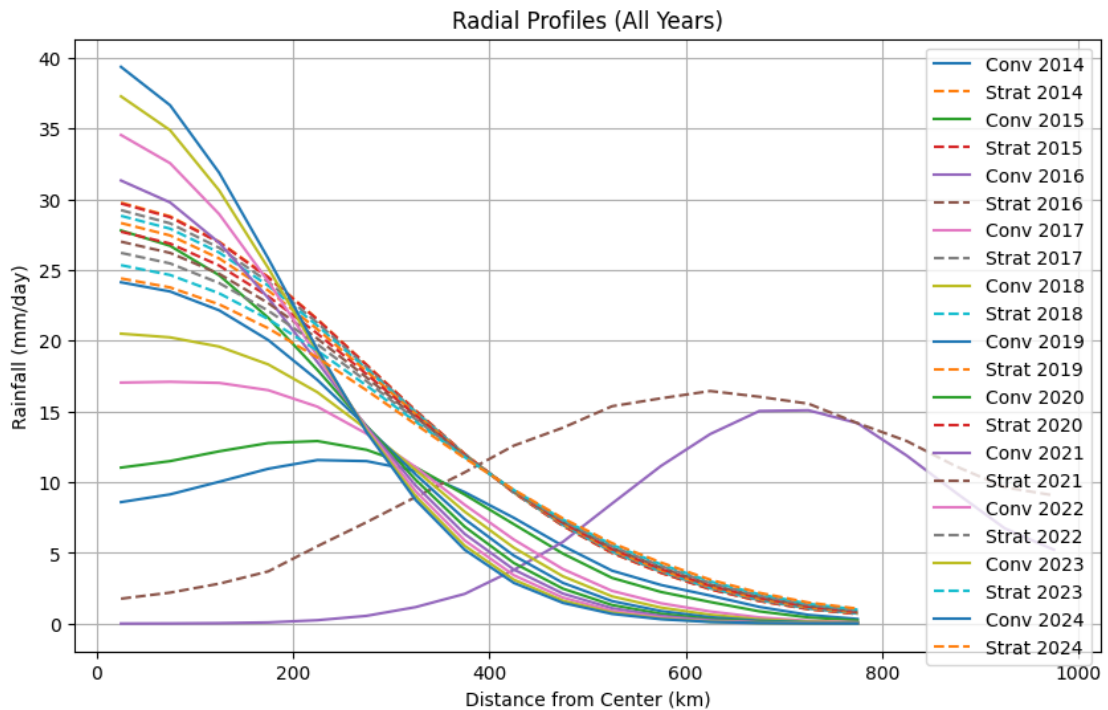


Figure 4.22: Annual radial distribution of convective rainfall (more than 20.0 mm/day) during JJAS (2014–2024) from MD centers.

Comparative Insights and Implications The radial composite analysis suggests that stratiform rainfall exhibits a more stable, symmetric distribution, while convective rainfall is highly asymmetric and spatially concentrated. This distinction is crucial for hydrological modeling, especially in predicting rainfall extremes and assessing flood risk zones.

Enhanced convective activity in recent years (2022–2024) may be indicative of increased vertical instability and warmer SSTs in the Bay of Bengal, consistent with recent findings (Roxy *et al.*, 2017; Sandeep *et al.*, 2020). Meanwhile, the spatial regularity of stratiform rainbands supports their attribution to organized systems rather than isolated convection.

4.7. Azimuthal Distribution of Convective and Stratiform Rainfall around Monsoon Depressions (JJAS 2014–2024)

This section presents an interannual analysis of the azimuthal variability of rainfall associated with monsoon depressions (MDs) during JJAS from 2014 to 2024. The goal is to assess directional asymmetries and annual shifts in rainfall organization with respect to MD centers. Figures 4.23 to 4.25 highlight comparative and quantitative patterns in azimuthal rainfall for both convective and stratiform components.

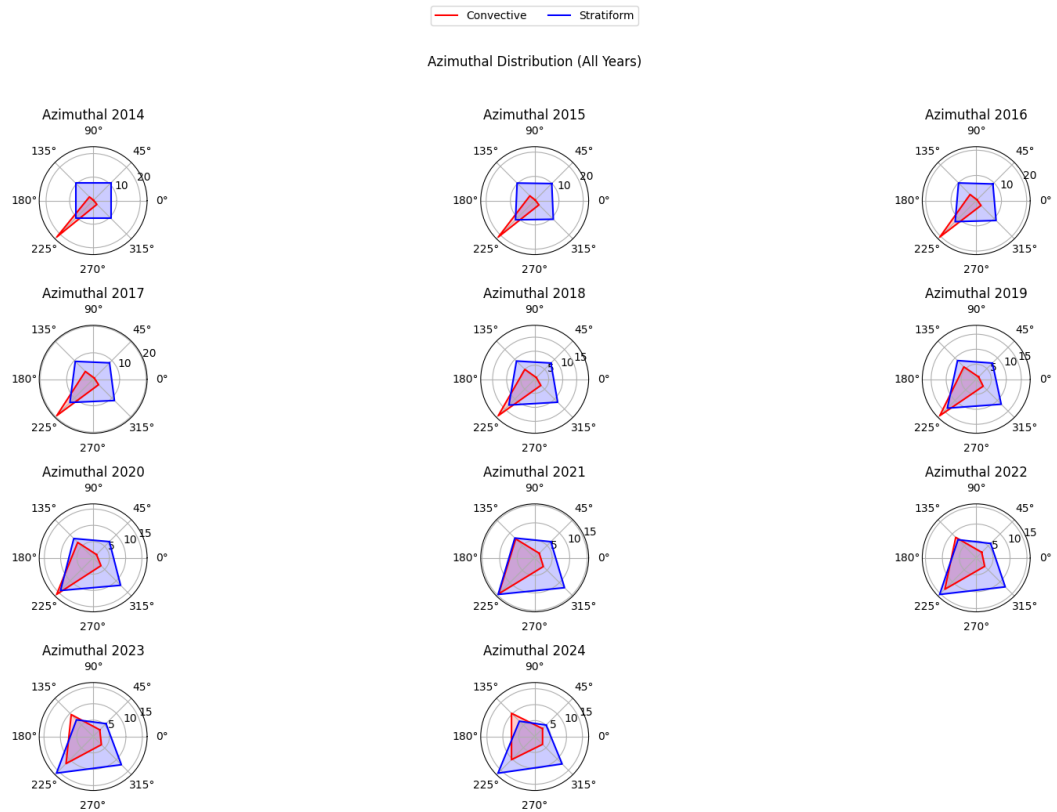


Figure 4.23: Radar plots showing azimuthal distribution of convective and stratiform rainfall around MD centers (2014–2024). Mean values are sectorally averaged over eight 45° bins.

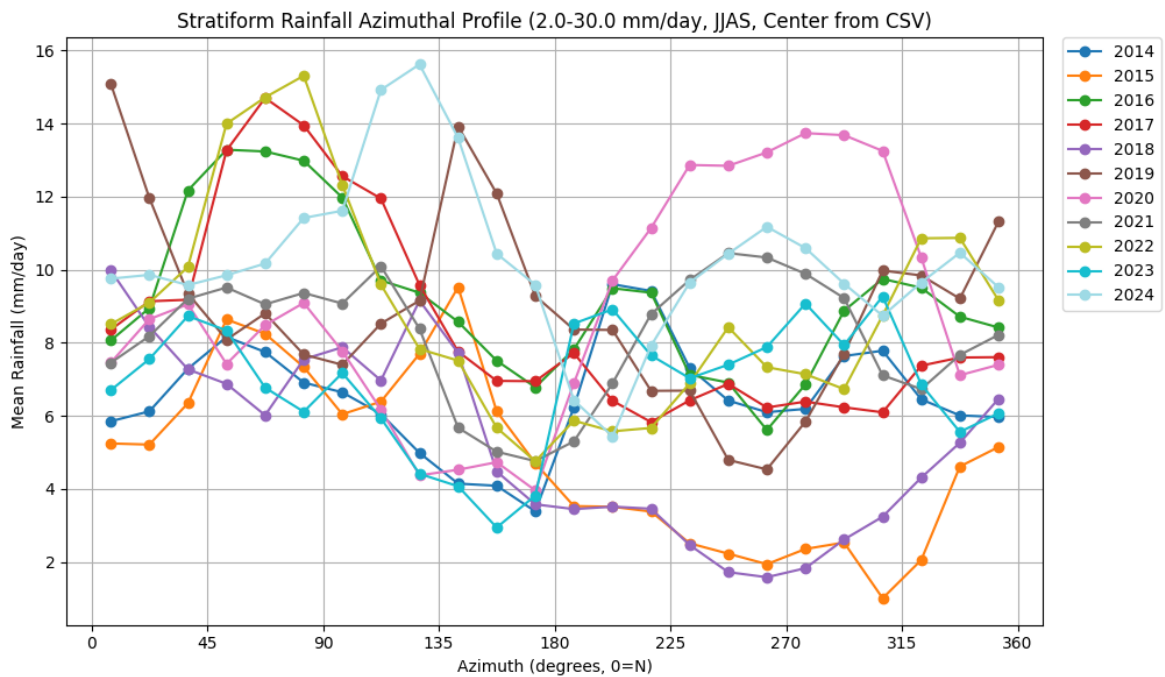


Figure 4.24: Stratiform rainfall azimuthal profiles (2.0–30.0 mm/day) across years. Each curve represents mean rainfall in 10° bins centered from MD tracks during JJAS.

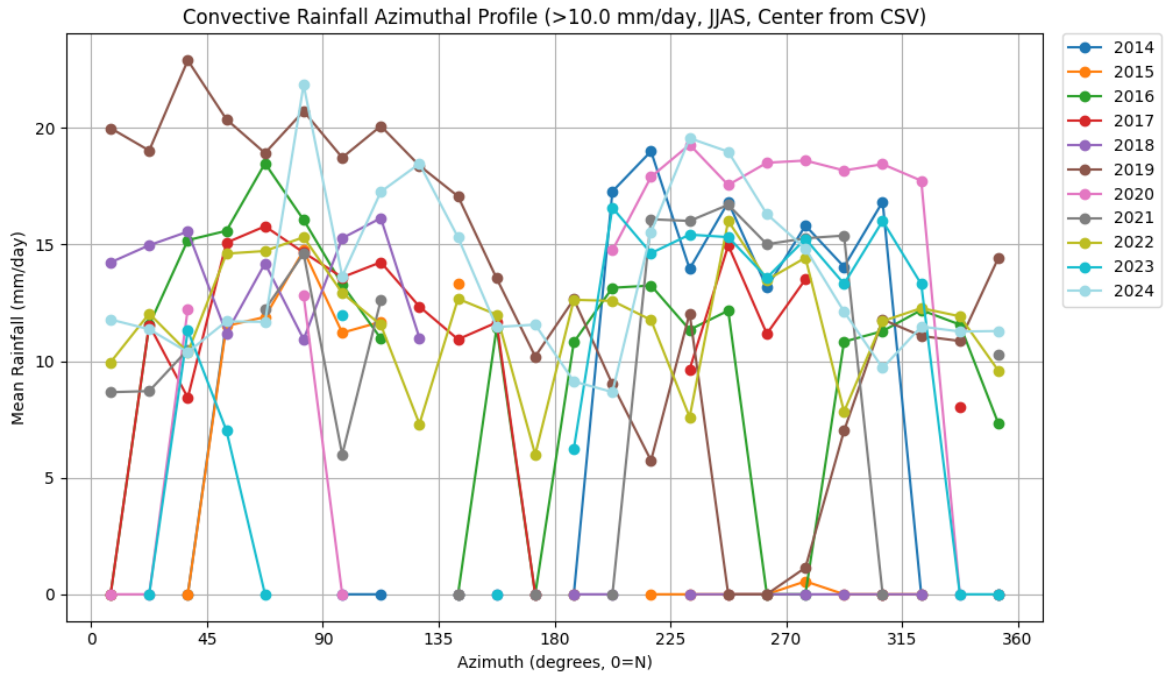


Figure 4.25: Convective rainfall azimuthal profiles (>10.0 mm/day) across years. Strong asymmetries are observed, especially during active MD years.

Sectoral Patterns and Dominance The radar plots (Fig. 4.23) show a general dominance of rainfall in the southeast (225°) and southeast (135°) quadrants, especially for stratiform precipitation. Convective rainfall, however, displays larger interannual variability and often peaks in isolated directions such as northwest (315°) or northeast (45°), highlighting the chaotic nature of embedded convection within MD structures.

Stratiform Rainfall Azimuthal Profiles Stratiform rainfall (Fig. 4.24) tends to be more azimuthally uniform, with smoother profiles and moderate interannual variation. Notable peaks above 14 mm/day occur during:

- 2016–2017: Suggestive of stronger trailing stratiform rainbands.
- 2020 and 2023: Widespread distribution with higher values in the southern sectors (180° – 270°).

Despite fluctuations, most years maintain rainfall more than 6 mm/day across a wide range of directions, strengthening the persistent stratiform mesoscale structure of MD.

Convective Rainfall Azimuthal Profiles In contrast, convective rainfall (Fig. 4.25) exhibits strong directional asymmetry with sharp peaks in specific quadrants, which vary significantly from year to year. High-intensity convective sectors include:

- 2016 and 2019: Peaks less than 20 mm/day in the northeast to southeast (45° – 135°) sectors.
- 2020 and 2023: Convective enhancement in western quadrants, possibly due to vortex tilt or shear alignment.

Zero or near-zero values in certain directions for many years highlight the episodic and localized nature of deep convection compared to stratiform spread.

Implications for Rainfall Forecasting These findings underline the importance of resolving azimuthal rainfall asymmetries in models. The consistent stratiform belt in southern quadrants provides a predictable structure, whereas the shifting convective maxima suggest the need for real-time monitoring and ensemble forecasts to capture severe precipitation risks.

5. Future Work

The present analysis opens multiple avenues for future investigation aimed at advancing the understanding, monitoring, and prediction of monsoon depression (MD) rainfall over the Indian subcontinent. The following directions are proposed:

- **Machine Learning-Based Rainfall Prediction:** Develop and implement ML/DL models (e.g., LSTM, CNN, or hybrid networks) for forecasting MD-associated rainfall using inputs such as SST anomalies, MJO indices, outgoing longwave radiation (OLR), and upper-air winds.
- **Tracking MD Lifecycle and Structure:** Use high-resolution datasets (e.g., ERA5, INSAT-3D, GPM IMERG) to extract spatial–temporal evolution of individual MD events, including vertical structure, propagation speed, and convective-stratiform transition zones.

- **Climate Attribution Studies:** Conduct detailed attribution experiments using coupled climate models (e.g., CESM, CFSv2) to isolate the specific roles of ENSO flavors (Modoki vs. canonical), IOD phases, and MJO phases in modulating MD frequency, tracks, and rainfall.
- **Impact-Based Risk Assessment:** Link MD rainfall anomalies to flood damage datasets, agricultural yield loss, and reservoir inflow patterns to assess socio-economic impacts. This can aid in creating early warning systems and climate-resilient infrastructure planning.
- **Regional Downscaling and Bias Correction:** Apply statistical and dynamical downscaling methods to correct biases in coarse-resolution GCM outputs, enabling reliable regional projections of MD rainfall under future warming scenarios.
- **Extension to Ocean–Atmosphere Coupling:** Study the interaction between Bay of Bengal ocean heat content, surface salinity, and MD genesis using ocean reanalysis products and coupled simulations to explore feedback mechanisms.

These directions aim to not only enhance scientific understanding but also bridge the gap between monsoon research and operational decision-making in the context of climate variability and extremes.

6. Conclusion

This study presents a comprehensive examination of convective and stratiform rainfall structures associated with monsoon depressions (MDs) over the Indian subcontinent during the southwest monsoon season (JJAS) spanning the period 2014–2024. Leveraging high-resolution satellite-derived rainfall data, in particular the GPM IMERG product, alongside ancillary reanalysis and gauge-based datasets (ERA5, MERRA-2, IMD gridded raingauge, and GPM DPR Level-2A), we evaluated the spatiotemporal variability, radial profiles, and azimuthal characteristics of MD-associated precipitation.

Among the datasets utilized, GPM IMERG Final Run daily precipitation data emerged as the most consistent, high-resolution, and spatially complete product for analyzing convective and stratiform rainfall dynamics over diverse physiographic regions in India. While ERA5 and MERRA-2 reanalyses provided valuable insights into large-scale circulation and vertical structure, and GPM DPR allowed microphysical validation at select overpasses, IMERG offered unmatched daily continuity, resolution ($0.1^\circ \times 0.1^\circ$), and skill in capturing both intense convective bursts and widespread stratiform rainfall. Thus, GPM IMERG data were selected as the primary basis for composite and statistical analyses in this research.

Radial distribution composites of rainfall centered on MD positions revealed distinct structural differences between convective and stratiform precipitation. Convective rainfall was characterized by highly asymmetric, localized cores typically within 100–400 km of MD centers, often varying in intensity and orientation year-to-year. In contrast, stratiform rainfall exhibited more uniform annular spread between 200–600 km, forming broad trailing rainbands that contributed substantially to overall rainfall volume. Azimuthal composites further highlighted the dominance of stratiform rainfall in the southwestern and southeastern quadrants, while convective maxima shifted directionally across years, likely influenced by vortex tilt, environmental shear, and moisture availability.

Temporal trends (2014–2024) suggested increasing convective rainfall over coastal and southern India, with simultaneous declines in central India, reflecting possible shifts in MD trajectories and thermodynamic environments. Stratiform rainfall trends, though more spatially neutral, also showed localized increases in orographically influenced zones (e.g., Konkan coast, northeast India), underscoring the importance of terrain interactions.

The integrated approach of combining satellite, reanalysis, and observational data has allowed for a robust understanding of rainfall asymmetries, intensification patterns, and sectoral contributions around MDs. These insights have critical implications for improving

quantitative precipitation forecasts (QPF), flood risk assessment, and water resource planning in monsoon-dominated regions. The evolving convective structures also warrant closer examination under climate change projections, especially with regard to increasing rainfall extremes and mesoscale system behavior.

In future work, GPM IMERG data will continue to form the foundation for extended research on diurnal rainfall cycles, microphysical evolution of convective systems, and coupling with land–atmosphere interactions during monsoon disturbances. Additional validation against localized raingauge networks and hydrological models will further enhance the application potential of these findings in operational forecasting and climate resilience planning.

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